



# Semantic segmentation assisted deep ensemble feature learning model for skin-cancer detection and classification: SDENet

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## Abstract

The last few years have witnessed rapid increase in skin cancer caused mortality rate. Despite innovations and growth in vision-computing and artificial intelligence technologies, the complex shapes, sizes, textural patterns and ambiguous edges limits the reliability of existing approaches. Nevertheless, unlike traditional approaches the deep learning methods have performed superior; yet, the demands for the superior skin-lesion segmentation, ROI-specific feature extraction and learning can't be ruled out. Moreover, it requires addressing class-imbalance problems as well to avoid skewed learning and prediction. Considering it as motivation, in this paper a novel and robust semantic segmentation assisted deep ensemble feature learning environment for skin-cancer detection and classification (SDENet) is proposed. The proposed SDENet model is targeted to perform multi-class skin-cancer classification. To achieve it, the SDENet at first performs standard pre-processing followed by synthetic minority over-sampling (SMOTE) to alleviate class-imbalance problem. Subsequently, it performs firefly heuristic algorithm based Fuzzy C-means clustering to segment skin-lesions (say, ROI), which is followed by ROI-specific deep spatio-textural ensemble feature extraction and fusion (DeS-TEFF). Specifically, SDENet makes use of the AlexNet deep network, DenseNet121 and Gray level co-occurrence matrix (GLCM) feature extraction methods. Here, AlexNet serves high-dimensional information rich features, while DenseNet121 yields layer-wise learning and feature reuse driven feature-set. Performing horizontal concatenation over the AlexNet, DenseNet121 and GLCM features, the principal component analysis (PCA) feature selection was performed, which helped to avoid local minima and convergence. The selected features were normalized by means of the z-score normalization so as to avoid over-fitting problems. Finally, the normalized features were trained and classified by using Heterogenous Ensemble Classifier, embodying SVM, DT, Random Forest, Extra Tree Classifier and XGBoost classifiers. The maximum voting ensemble-based classification over HAM10000 dataset exhibited the average accuracy of 98.97%, precision 99.38%, recall 98.94% and F-Measure 0.99, confirming its superiority over other existing approaches for real-time skin cancer diagnosis purposes.

**Keywords** Skin cancer detection and classification · Deep ensemble features · Skin-lesion segmentation · Ensemble classifier · Computer-aided diagnosis

## Abbreviations

|          |                                                                       |
|----------|-----------------------------------------------------------------------|
| DeS-TEFF | Deep Spatio-Textural Ensemble Feature Fusion                          |
| ETC      | Extra Tree Classifier                                                 |
| FCM      | Fuzzy C-Means                                                         |
| FFCM     | Firefly-Optimized Fuzzy C-Means                                       |
| FA       | Firefly Algorithm                                                     |
| GLCM     | Gray Level Co-occurrence Matrix                                       |
| HAM10000 | Human Against Machine with 10000 Training Images Dataset              |
| ROI      | Region of Interest                                                    |
| SDENet   | Semantic Segmentation Assisted Deep Ensemble Feature Learning Network |

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|       |                                            |
|-------|--------------------------------------------|
| SMOTE | Synthetic Minority Over-sampling Technique |
| UNet  | Universal Network for Image Segmentation   |

## Introduction

The last few years have witnessed rapid global increase. The severity of skin cancer can be considered by a recent study which states that the skin-cancer accounts almost 33.33% of the total cancer cases [1]. Considering such serious health concern, recently the World Health Organization (WHO) too declared skin-cancer as a rising global concern [1]. A recent study indicates that despite economic prosperity and improved living standards, the economies like United States, Canada, and Australia have witnessed more than 15,000 deaths every year [2]. Global health bulletin indicates that almost 7180 people died in the United States alone because of a single type of skin cancer in the US in 2021, while this rate is supposed to grow over 7650 in the subsequent year [2]. The global surveys also reveal that almost 0.1 million melanoma skin-cancers and 3 million non-melanoma skin cancers are diagnosed globally each year [1]. The studies reveal that the continuous depletion of the ozone layer has been increasing hazardous ultraviolet radiation on the earth surface that consequently has been causing skin-cell damages and hence skin cancer [3]. On the other hand, the changing lifestyle such as the alcohol intake, smoking, infections, local hazardous chemical impacts, pollution etc. are the decisive reasons behind increasing skin cancer cases or skin's malignancy [3]. Typically, skin cancers are categorized into two broad types; malignant and non-malignant, where the first type can be of the different shapes and sizes, and malignant in nature [4]. Additionally, there exists other different types of skin cancers such as the malignant cutaneous adnexa, fibrous histiocytomas, Kaposi's sarcoma, and pleomorphic sarcoma. Moreover, the squamous cell cancer (SCC), basal cell cancer (BCC), malignant melanoma, and dermatofibrosarcoma represent the key types of malignant skin cancers [1]. Though, there exist some other types of skin cancers as well, which are the malignant cutaneous adnexa, fibrous histiocytomas, Kaposi's sarcoma, and pleomorphic sarcoma. The clinical assessments reveal that the malignant melanoma is more harmful due to its ability called metastatic that spreads it across the body's regions [4]. The uncontrolled growth in malignant cells often results into the high mortality rate and hence earlier detection and identification of the skin cancer is inevitable to reduce mortality rate by diagnosing patients effectively on-time [5].

Traditionally, the medical experts examine skin lesions shape, size, color and allied textural patterns to detect and diagnose skin cancer; however, the likelihood of human

error can't be ruled out. Moreover, the diversity of the aforesaid lesion's characteristics over the different demographics or patients, guaranteeing reliability of the traditional diagnosis measure is difficult. To alleviate such challenges, in the last few years computer-vision based solutions, often called computer-aided diagnosis (CAD) has gained widespread attention [6]. The high pace growth in the vision technologies, artificial intelligence (AI) and machine learning (ML) has broadened the horizon for skin cancer detection, localization and classification [7]. Learning over the different spatio-textural cues and deep features from the dermatological images, the AI techniques can be designed to perform automatic skin-lesion detection and cancer classification. Undeniably, the major innovations, the deep learning and ML techniques have performed superior towards vision-computing based skin-cancer classification. Despite this the traditional vision-based approaches exploiting size, shape, texture, color etc. undergo reduced performance, especially over scalable application environment [7]. As stated above, the decisive diversity over skin lesions, skin types, local imaging conditions and patient's ethnicities limit the efficacy of the major at hand solutions [7]. Merely exploiting above stated handcrafted features (i.e., size, shape, color, etc.) can yield false positive performance due to their inability to distinguish a few malignant and benign images, especially attributed due to their corresponding ambiguous or near-similar dermatologic manifestation. In real-world diagnosis environment, the nodules with the pearly-rolled edges and skin-colored conditions on face can be mistakenly be classified as the BCC. On the contrary, the other skin neoplasm can be SCC, amelanotic melanoma, and trichoepithelioma. On the other hand, the SSC, which resembles actinic keratosis (AK), amelanotic melanoma, BCC, warts, spindle cell tumor, traumatic wounds, or other benign tumors [5]. In the same manner, the melanoma, which attributes the highest mortality rate [5], can pose difficulty to detect and classify accurately, due to the resembling characteristics with seborrheic keratosis, melanocytic nevus, pigmented BCC, lentigo, angiokeratoma, dermatofibroma, etc [8]. In this case, merely applying local handcrafted features can't guarantee reliability towards clinical decisions. Ironically, considering aforesaid challenges a large number of existing efforts have performed binary classification to either classify benign and malignant images or the melanoma and benign. On the contrary, a real-world solution demands the ability to detect, segment and classify the different types of skin cancers so that the appropriate clinical decisions could be made to reduce mortality rate.

To achieve higher accuracy towards the skin-cancer classification, the efforts have been made to improve the different associated functions such as the skin-lesion segmentation, feature extraction and classification. The accurate detection

and segmentation of the skin lesions can enable extracting region-of-interest (ROI) specific feature extraction that eventually can help to learn fine grained details of each skin-cancer types and hence can enable higher accuracy [9]. On the other hand, extracting sufficiently large and intrinsically information rich features can help to train the models effectively to improve classification accuracy [10]. The machine learning algorithms on the other hand focus to learn over the highly discriminative feature sets to achieve accurate skin-cancer classification. Yet, identifying an optimal set of segmentation method, feature extraction and classification environment remains an open challenge for academia-industry. On the other hand, poor performance by the major multi-class classification models classifying dermatological images into the different categories triggers researchers to develop more efficient and robust CAD solution. It can be considered as the key driving force behind this research.

Contemporarily, the major state-of-arts applying vision computing and dermoscopy images [10], which mainly exploit the morphological features pertaining to the skin lesions or nodules [11], yield average accuracies in the range of 65% -75% [12]. Such poor accuracy confines their suitability towards real-world diagnosis tasks [12]. Though, to improve accuracy AI techniques have played decisive role which help to segment skin-lesions accurately in more affordable manner [13]. Yet, the lack of features and classification environment can confine accuracy and reliability. To address this problem, in the past different deep learning models are proposed including the convolutional neural networks (CNNs) [14], deep neural network (DNN), residual networks like ResNet [7, 15], DenseNet [7], transfer learning models [16] and other variants like SqueezeNet, VGG-Net [7], MobileNet [8, 17], EfficientNet etc. These deep models mainly focus on extracting deep features from the different types of the dermoscopic images, which are trained over the machine learning methods to perform skin-cancer classification. However, these deep models undergo different challenges such as the lack of contextual details, gradient vanishing, gradient explode and accuracy degradation. For instance, the CNN models and its derived deep learning methods often exploit local hierarchical features and lack contextual details or the global features, which can have the decisive impact on the learning and prediction results. On the other hand, the CNNs or DNNs applying larger number of convolutional layers often undergo gradient vanishing. Though, ResNet models [7, 15] have performed layer-wise learning and feature reuse to alleviate aforesaid gradient vanishing problem; however, can't guarantee the optimality as standalone feature. A few recent literatures indicate that the ensemble of deep features can yield superior results than the standalone deep models [18]. A few deep-based image classification problems have shown that the strategic

amalgamation of the residual learning with other high-dimensional features can improve learning, while alleviating gradient vanishing and gradient explode problems [7, 15]. In addition to the above stated challenges and allied scopes, in image processing-based CAD solutions, class-imbalance has always remained a challenge. This is mainly because of the imbalanced or uneven image samples [6]. The data imbalance can exist on both instance-level as well as the sample level, where in the earlier case the ROI can be extremely smaller in size. On the other hand, sample-level imbalance can be due to the imbalance data samples pertaining to the different types of the skin-cancer (dermatological) images [17, 18]. Extracting and learning features from such imbalanced images (skewed distribution) can impact learning efficiency or can result skewed prediction. Consequently, it can impact overall learning and prediction results [17]. To address such issues, data resampling, (image) augmentation methods [18] have been proposed in the past. Additionally, to improve learning and prediction efficiency, different measures such as the training data transformation, classifier ensemble-based predictions are proposed. Yet, the likelihood of over-fitting with such methods can't be ruled out [17].

Considering above discussed inferences and allied scopes this paper proposed a novel and robust semantic segmentation assisted deep ensemble feature learning environment for skin-cancer detection and classification (SDENet). The SDENet performed multi-class skin-cancer classification. The SDENet initially performs pre-processing followed by synthetic minority over-sampling (SMOTE) to alleviate class-imbalance problem. Subsequently, it performs firefly heuristic algorithm based Fuzzy C-means clustering to segment skin-lesions, which is followed by ROI-specific deep spatio-textural ensemble feature extraction and fusion (DeS-TEFF). Specifically, SDENet makes use of the AlexNet deep network, DenseNet121 and Gray level co-occurrence matrix (GLCM) feature extraction methods. Here, AlexNet serves high-dimensional information rich features, while DenseNet121 yields layer-wise learning and feature reuse driven feature-set. Performing horizontal concatenation over the AlexNet, DenseNet121 and GLCM features, the principal component analysis (PCA) feature selection was performed, which helped to avoid local minima and convergence. The selected features were normalized by means of the z-score normalization so as to avoid over-fitting problems. Finally, the normalized features were trained and classified by using Heterogenous Ensemble Classifier, embodying SVM, DT, Random Forest, Extra Tree Classifier and XGBoost classifiers. The maximum voting ensemble-based classification over HAM10000 dataset exhibited the average accuracy of 98.97%, precision 99.38%, recall 98.94% and F-Measure 0.99, confirming its superiority over

other existing approaches for real-time skin cancer diagnosis purposes.

Motivated by the challenges identified in the literature, the present study proposes a novel framework, termed Semantic Segmentation Assisted Deep Ensemble Feature Learning Model (SDENet), for accurate and interpretable skin-cancer detection and classification. The novelty of SDENet lies in its integration of semantic segmentation, deep spatio-textural feature fusion, and heterogeneous ensemble learning within a single end-to-end environment.

Unlike most conventional CNN-based or transfer learning models, which often suffer from class imbalance, redundant feature learning, or poor ROI localization, SDENet introduces several key innovations:

- A Firefly-optimized Fuzzy C-Means (FFCM) segmentation approach for automatic and precise lesion extraction, reducing dependence on manual or threshold-based segmentation.
- A Deep Spatio-Textural Ensemble Feature Fusion (DeS-TEFF) strategy that integrates AlexNet, DenseNet121, and GLCM features to capture both global and fine-grained texture patterns.
- A Principal Component Analysis (PCA) based feature selection scheme followed by a heterogeneous ensemble classifier (SVM, DT, RF, ETC, XGBoost) with maximum voting for robust multiclass classification.

Collectively, these contributions make SDENet robust against overfitting, enhance generalization, and improve interpretability while achieving superior diagnostic accuracy compared to state-of-the-art methods on the HAM10000 dataset.

The remainder of this paper is structured as follows. Section II presents related studies on skin lesion segmentation and classification. Section III defines the research questions that guide this work. Section IV details the proposed SDENet methodology, including data preprocessing, FFCM-based segmentation, feature extraction, and ensemble learning. Section V reports the experimental setup, results, and comparative performance analysis. Finally, Section VI concludes the study with key findings and directions for future research.

## Related work

Zia et al. [19] appended MobileNet and DenseNet201 with an additional convolutional layer to perform skin cancer detection, which achieved the highest (binary) classification accuracy of 95.50%. Though, MobileNetV2 yielded the highest accuracy of 91.86% on the ISIC skin cancer

dataset. Gouda et al. [20] applied deep-CNN (DCNN) to perform binary skin cancer classification (i.e., benign and malignant). Simulations confirmed that the ResNet50 and InceptionV3 model yield the prediction accuracies of 83.7% and 85.8%, correspondingly on the ISIC 2018 dataset. Dorj et al. [21] applied AlexNet model as feature extraction tool, which was followed by the support vector machine (SVM)-based classification to perform skin cancer diagnosis. It achieved the highest accuracy of 95.1%, while the sensitivity and specificity were measured as 98.9% and 94.17%, respectively. Fu'adah et al. [22] applied CNN with the different learning models like RMSprop, SGD and Adam, where the ADAM-based CNN yielded the highest skin cancer classification accuracy of 83% using ISIC dataset. Barata et al. [23] applied CNN model for feature extraction and cross-entropy-based classification, where the simulation results over ISIC 2017 datasets resulted the balanced accuracy of 65.5%. Nigar et al. [24] on the other hand applied ResNet-18 deep model for skin lesion classification that resulted the classification accuracy of 94.47%, precision of 93.57%, recall of 94.01%, an F1 score of 94.45% on ISIC 2019 dataset. Ameri et al. [25] too applied AlexNet-CNN deep learning model for skin cancer diagnosis by applying the HAM1000 dermoscopy images. It performed two-class classification and annotated each image as non-melanoma and melanoma, where the highest ROC curve area obtained was measured as 0.91. Kumar et al. [26] applied CNN to diagnose skin cancer. Training over the ISIC dataset, embodying 2637 images, it resulted the highest accuracy of 88%. Nawaz et al. [27] applied the DenseNet77 and UNet deep models, where the first was used as the encoder for the UNet to extract high-representative features. Segmenting lesions by using UNet, the features were trained with the cross-entropy objective functions, which resulted the skin cancer classification accuracies of 99.21% and 99.51% in ISIC-2017 and ISIC-2018, correspondingly. The authors claimed that the use of Unet semantic segmentation enabled superior feature for learning and prediction, which helped to improve accuracy. Maqsood et al. [28] designed a custom 26-layer CNN for skin lesion segmentation, which was followed by feature extraction by using XceptionNet, ResNet50, ResNet101, and VGG16 deep models. Their extracted features were fused to yield composite feature, which was later processed for feature selection by applying Poisson distribution. Training the extracted features by using SVM, it resulted the accuracies of 98.57%, 98.62%, 93.47%, and 98.98% on HAM10000, ISIC2018, ISIC2019, and PH2 datasets, respectively. Abayomi-Alli et al. [29] at first applied Synthetic Minority Oversampling Technique (SMOTE)-based resampling to alleviate class-imbalance problem, which was followed by the SqueezeNet-based feature extraction and learning. This model yielded the highest



skin-cancer classification accuracy of 92.18%, sensitivity, specificity and F-Measure values of 80.77%, 95.1%, and 80.84%, correspondingly for binary classification tasks. For multi-class classification problem, it resulted the highest accuracy of 89.2%. Jiahao et al. [30] assessed VGG-16, ResNet50, and EfficientNet-B5 on ISIC 2020 dataset for skin cancer diagnosis. They concluded that the EfficientNet-B5 yields superior results. Zhang and Wang [31] found that the DenseNet201 outperforms VGG16 and ResNet50 on the ISIC 2020 dermoscopic dataset for skin-cancer classification. Zhang et al. [32] developed Whale optimization-based CNN deep model for skin cancer detection.

Liu et al. [33] on the other hand designed ResNet-based clinical-inspired network (CI-Net), which was assessed for skin cancer detection on the ISIC 2016–2020 and PH2 datasets. Kaur et al. [34] developed LCNet, a CNN variant with the image pre-processing, oversampling, and augmentation to improve skin-cancer diagnosis results on ISIC 2016, ISIC 2017, and ISIC 2020 datasets. Reis et al. [35] proposed InSi-Net, specially designed with segmentation driven deep-models including Inception-v3, DenseNet-201, ResNet-152v2, and EfficientNet-B0. Yet, the poor accuracies confine its real-time significance. López-Labraca et al. [36] designed CNN-enabled deep learning environment that enables neurons to detect and learn visual features and assess activations to extract and learn intrinsic features towards skin-cancer diagnosis. Rezk et al. [37] developed a DNN-based interpretable skin cancer diagnostic model. Ensuring its scalability over real-world environment and diverse data inputs seems difficult. In [38], the authors at first resampled skin lesion images, which was followed by the Unet segmentation and DenseNet-based feature extraction. The average accuracy measured was 84%. A similar effort was made in [39], where pre-processing the input lesion images, segmentation was done. More specifically, the noise elements including gels, bubbles, hair etc. were removed in pre-processing. The authors applied CNN deep model to perform feature extraction [40]; yet, it could achieve the highest accuracy of 78%. They proposed fully convolutional neural network for semantic pixel-by-pixel segmentation. In [41] as well, CNN was applied as a feature extraction model, which was additionally armored with a pixel-wise classification layer to train the model [42]. Being designed as VGG-16's 13 convolution layers it achieved the highest accuracy of 88% on HAM10000 dataset. In [43], the authors made use of the ABCD rule, GLCM, and HOG features to perform skin cancer classification. Training aforesaid features with SVM classifier, it achieved the highest accuracy of 97.8%. Though, k-NN classifier resulted the sensitivity was 86.2% and the specificity was 85%. K-Means clustering method was applied in [44], which resulted the accuracy of 52.63% for skin cancer classification. Though, back-propagation

neural network (BPNN) resulted the accuracy in the range of 60% to 75% over the different sets of input images. In [45], the authors assessed the relative performance by the different deep networks like Inception V3, ResNet and VGG 16 and MobileNet for skin cancer classification. The authors [46] applied MobileNet and VGG-16 models to perform skin cancer classification using HAM10000 dataset. The MobileNet and VGG-16 [47] models resulted the highest skin-cancer prediction accuracy of 81.52% and 80.07%, respectively. The authors [48] developed RetinaNet-based skin-cancer classification. several different regions. They also applied conditional random field (CRF) to localize the segmented lesion regions. Executing their model over Pedro Hispano (PH2) and ISIC 2018 datasets, it resulted sensitivity and specificity of 93.20% and 97.70%, correspondingly. The accuracy measured was 94.20% on ISIC 2018 dataset. In [49], a 38-layer deep network was developed that over the HAM10000 and ISIC2019 datasets exhibited the accuracy of 94.45% and 93.06%, correspondingly. In [50] Unet was applied for lesion segmentation, while LinkNet a transfer learning model was used to extract feature and perform classification. On the datasets ISIC 2018 dataset, it exhibited the Dice value of 89.3. The authors [51] designed an explainable CNN driven stacked ensemble network for melanoma detection. The proposed ensemble deep network resulted the melanoma cancer prediction accuracy, sensitivity and AUC of 95.76%, 96.67% and 95.7%, correspondingly.

In [52], at first canny edge detector was applied to segment skin lesions, which was followed by the ABCD rule-based feature extraction. It could achieve the highest accuracy of 93.6%, precision 100%, recall 92.5%, and F-measure of 96.1%. In [53], Otsu-threshold based segmentation was done to detect skin lesions, which was followed by SIFT, skewness features. Despite an exhaustive computation, it could achieve the highest accuracy of 80.00%. In [54], ResNet50 deep features were obtained from the dermoscopy images, which were trained and classified for skin-cancer classification. It achieved the highest AUC of 87.00%, while accuracy measured was 94.00% on ISIC dataset. The HOG-based features resulted the highest specificity and sensitivity of 97.00% and 62.00%, respectively. On the other hand, in [55], the different features including HOG, SURF were obtained from the graph-cut segmentation outputs. It yielded the highest accuracy of 95.86% on PH2 dataset, while resulted the accuracy of 94.92% on ISIC 2017 dataset. The CNN features were trained over SVM classifier in [56], which yielded the highest accuracy of 82.6% on ISIC dataset. The authors [57] extracted GLCM features from the segmented skin-lesions, which was later processed for principal component analysis (PCA) feature selection. Training over SVM classifier it yielded the highest accuracy of 92.10%. On the other hand, the local binary

pattern (LBP) trained over SVM classifier yielded the highest accuracy of 93.1% and specificity of 92.80%. In [58], at first OTSU-based segmentation was performed, which was followed by gradient histogram feature extraction and SVM-based classification, which showed the accuracy of 96.30%. On the bag-of-features obtained from the local feature extraction models, it resulted the highest skin-cancer detection accuracy of 87%. With the OTSU-segmented lesions, the authors [59] extracted shape, color, and texture features, which were trained over the decision tree (DT) classifier, which yielded the highest accuracy of 85.00%. Fuzzy-assisted OTSU segmentation was applied in [60] for skin lesion segmentation, which was followed by handcrafted feature extraction and learning. Despite their claim to have adaptive thresholding-based segmentation its use over non-linear data conditions seems suspicious. Exploiting shape, color, texture and the islands of color information, the ANN learning method resulted the highest accuracy of 94%. In [61], color features were trained over k-NN classifier which classified input images as benign, dysplastic, and malignant. It showed the specificity and sensitivity of 93.00% and 87.00%, correspondingly. In [62], the authors proposed CNN with illumination normalization method to improve skin-cancer classification results. Training over MobileNet and SqueezeNet models, it achieved the highest accuracy of 99.13% on MNISU HAN10000 dataset. In [63], fast fuzzy c-means (FCM) was applied to segment skin lesions from dermoscopic images. Subsequently, it generated superpixels from the images using MMLVR-WT, which was followed by the super-pixel images' histogram estimation to support FCM that resulted the sensitivity and specificity of 95.76% and 98.96%, correspondingly.

Tan et al. [64] applied particle swarm optimization (PSO) heuristic-based K-Means clustering algorithm for skin lesion segmentation, which was followed by CNN-based feature extraction and learning to perform skin-cancer classification (i.e., melanoma and nevus lesions). Kwasigroch et al. [65] designed CNN with the help of a hill climbing concept to reduce search space so as to improve learning and classification accuracy towards skin lesions classification. Adegun et al. [66] developed an encoder and decoder model with skip connection. CNN was used to segment skin lesions, which was followed by pixel-wise classification towards skin-cancer diagnosis. Song et al. [67] too applied CNN to segment skin lesions and classify it for skin-cancer diagnosis by exploiting Jaccard distance and focal loss values. Lequan et al. [68] developed deep CNN for melanoma detection, where the use of fully convolutional residual network with 16 residual blocks helped to improve segmentation and allied learning outputs. The SVM classifier resulted the accuracy of 85.5%. Noticeably, they found that the segmentation-based learning performs superior (85.5%), while

without segmentation it performs 82.8%. The authors [69] proposed a multi-scale CNN trained on an ImageNet by using an Inception-v3 DNN. Mahbod et al. [70] extracted deep features by using AlexNet, ResNet-18, and VGG-16 deep models to classify skin cancer. The extracted features from SVM classifier yielded AUC of 97.55%. Kalouche et al. [71] proposed a VGG-16 pretrained deep CNN model resulted an accuracy of 78% for melanoma cancer detection. Hasan et al. [72] applied CNN model to classify skin-cancer that achieved accuracy of 92.2% on the ISIC 2018 dataset. Shorfuzzaman et al. [73] used CNNs that achieved classification accuracy of 94.87% on ISIC 2019 dataset. Benyahia et al. [74] fused the deep features obtained from the VGG16, ResNet50, and Inception-v3 that achieved the accuracy of 88.1% on ISIS 2018 dataset. Agyenta et al. [75] trained CNN features from the ISIC 2019 dataset by using SVM, k-NN, Naïve Bayes (NB), random forest (RF) and gradient boosting classifiers. Yet, the highest accuracy measured was 94.52%. Bhosale et al. [76] applied MobileNet feature, trained over the RF classifier for skin-cancer classification that with PyTorch implementation yielded the highest accuracy of 86.5%. Similarly, CNN features trained over the different machine learning methods yielded the accuracy of 88% in [77] and 89.5% in [78]. Afza et al. [79] developed DNN ensemble feature with the extreme learning machine (ELM) to perform skin-cancer classification by using HAM10000 and ISIC2018 datasets, which yielded the highest accuracies of 93.40% and 94.36%, correspondingly. Popescu et al. [80] assessed the efficacy of the deep models including GoogleNet, ResNet-50, DenseNet201, InceptionResNet-V2, XceptionNet, MobileNet V2, ResNet-101, and AlexNet to perform skin-cancer classification. Zhang et al. [31] used the DenseNet deep model to detect and classify skin-cancer on the ISIC2020 dataset that resulted an AUC score of 0.925. Singha et al. [81] assessed efficacy of ResNet, VGG16, and MobileNetV2 deep models to perform skin-cancer classification on the ISIC 2020 database, which showed the accuracies of 98.73% to 99.76%, and 98.39%, correspondingly. Keerthana et al. [82] with the CNN features on the ISBI 2016 dataset exhibited the accuracy of 88.02%. Naeem et al. [83] developed SCDNet, which was designed by using VGG16 with the CNN model to perform skin cancer classification on HAM10000 dataset. Amongst the assessed deep models, the XceptionNet showed the highest accuracy (90.48%).

Zhang Z., et al. Enhanced Deep Information Fusion based on Residual U-Net (ED-ResUNet) has shown that introducing residual links and multi-scale fusion can significantly improve segmentation accuracy while keeping the model lightweight [84]. Likewise, Liu et al. presented the MRP-UNet, a multiscale input fusion network capable of preserving both global and local lesion details, leading to

more reliable segmentation outcomes on dermoscopic datasets [85].

Alongside segmentation, researchers have also focused on improving classification accuracy through ensemble-based learning. Rather than relying on a single deep network, several recent works have demonstrated that combining multiple models produces more stable and accurate predictions. The Double-AMIS ensemble model, for instance, integrated adaptive momentum and intelligent selection mechanisms to enhance skin cancer classification performance [86]. Similarly, Hosseinzadeh et al. introduced a deep ensemble approach that fused multiple CNN architectures to improve generalization on datasets such as HAM10000 and ISIC [87]. In a broader investigation, Chiu et al. provided evidence that ensemble learning consistently achieves better robustness and interpretability than standalone deep models, particularly in multiclass classification settings [88].

Another area that has recently drawn attention is the use of metaheuristic optimization algorithms for image segmentation and clustering. The Firefly Algorithm (FA), in particular, has been widely explored for tuning model parameters and enhancing segmentation precision. As reviewed by Li, Firefly-based optimization methods are valued for their simplicity and strong convergence behavior in complex search spaces [89]. More recently, Thomas et al. applied a Firefly-optimized Fuzzy C-Means (FFCM) clustering method to medical image segmentation and demonstrated improved accuracy compared to conventional FCM [90]. These findings support the design choice in our study to employ FFCM for automatic, ROI-specific segmentation before feature extraction. By integrating this segmentation strategy with deep ensemble feature fusion (DeS-TEFF) and a heterogeneous voting-based ensemble classifier, our proposed SDENet framework aims to achieve more reliable and explainable multiclass skin cancer detection and classification results.

Despite significant progress, most existing deep learning and ensemble models for skin cancer detection face two persistent challenges: class imbalance and limited feature diversity. Several studies have shown strong accuracy on balanced datasets, but their performance often declines when applied to real-world clinical data with underrepresented lesion categories. Moreover, many existing architectures rely exclusively on deep convolutional features, overlooking textural or statistical cues that could enhance class separability. To address these limitations, the proposed **SDENet** framework incorporates SMOTE for class rebalancing, FFCM-based ROI segmentation to ensure focus on diagnostically relevant regions, and a **Deep Spatio-Textural Ensemble Feature Fusion (DeS-TEFF)** strategy that combines deep and handcrafted descriptors to improve robustness and interpretability.

## Research questions

In sync with the overall research goals, allied technical scopes and the methodologies, this work defines certain (research) questions. The justifiable answers to these questions can put foundation for a robust and efficient skin-cancer detection and classification solution. These research questions are:

**RQ1** *Can the strategic amalgamation of the SMOTE resampling method and Firefly heuristic based ROI segmentation alleviate class-imbalance and skewed learning problem towards skin-cancer detection and classification?*

**RQ2** *Can the use of ROI-specific deep spatio-textural feature ensemble and fusion model (DeS-TEFF) be effective with the PCA feature selection and z-score normalization to yield highly accurate and reliable skin-cancer classification?*

**RQ3** *Can the use of heterogenous ensemble learning environment encompassing SVM, DT, RF, ETC and XGB for maximum voting-based classification be more effective and reliable than the standalone classifier driven CAD solution (for skin-cancer classification)?*

**RQ4** *Can the amalgamation of the RQ1, RQ2 and RQ3, yield a robust and efficient vision-based skin-cancer detection and classification system for run-time significance?*

Thus, the overall research intends to achieve answers for these key questions that eventually can put foundation for a robust skin-cancer detection and classification solution.

## System model

The proposed method encompasses the following steps:

1. *Data Acquisition and Preparation.*
2. *Pre-processing.*
3. *FFCM Semantic Segmentation for Skin Lesion (ROI) Detection and ROI-Specific RGB Conversion.*
4. *ROI-Specific Deep Spatio-Textural Feature Ensemble and Fusion.*
5. *Principle Component Analysis (PCA).*
6. *Z-Normalization, and.*
7. *E2E Maximum Voting Ensemble based multiclass Skin-Cancer Classification.*

The implementation schematic and allied algorithmic realization is depicted in Fig. 1. The detailed discussion of the overall proposed model is given in the subsequent sections.

#### A. Data Acquisition and Preparation.

In this work, HAM10000 dataset was considered, which is a well-known and extensively used medical imaging dataset for dermatology research. It is publicly available and can be found from the open access link [91]. The considered dataset embodies a total of 10,015 dermatological images representing pigmented skin lesions. These images are collected from the skin cancer patients and their records available at the Medical University of Vienna. This dataset is prepared to support input medical imaging resource for varied CAD tasks including skin lesion classification, skin-cancer detection and classification. The dataset embodies an array of lesions types with corresponding clinical features. The dataset embodies the dermatological images pertaining to the seven categories. A brief of these skin lesions is given as follows:

- *Actinic Keratosis (AK).*

It represents the rough, scaly patches on human skin surface. In case, it is not treated on time, it might grow further and can result into squamous cell carcinoma (SCC) that can further be increased due to the continuous exposure to the sun-light.

- *Basal Cell Carcinoma (BCC).*

In appearance, the BCC patches are very small, and often look like shiny bumps or nodules. Though, with the gradual and untreated state can turn out to be regionally locally invasive. It is one of the highly frequent types of skin cancer, diagnosed globally.

- *Benign Keratosis-like Lesions (BKL).*

Typically, benign skin (cancer) development or growths takes place in older adults. It embodies an array of lesions including solar lentigines and seborrheic keratosis.

- *Dermatofibroma (DR).*

It is a kind of benign fibrous tumor, which is often found on the legs, is often called dermatofibroma. Physically, it represents a uniform (in shape) and reddish-brown lesion growth.

- *Melanoma (MEL).*

It is a severe and life-threatening skin cancer case which is typically emerges from the Melanocytic. The alarming rate of melanoma and resulting mortality cases triggers academia-industries to develop a robust CAD solution which could detect and classify it to enable earlier diagnosis decisions.

- *Nevus (NV).*

Such skin-cancer cases are also known as moles, nevi, which can be benign lesions can be flat (uniform) and raised in appearance but do vary in color and size.

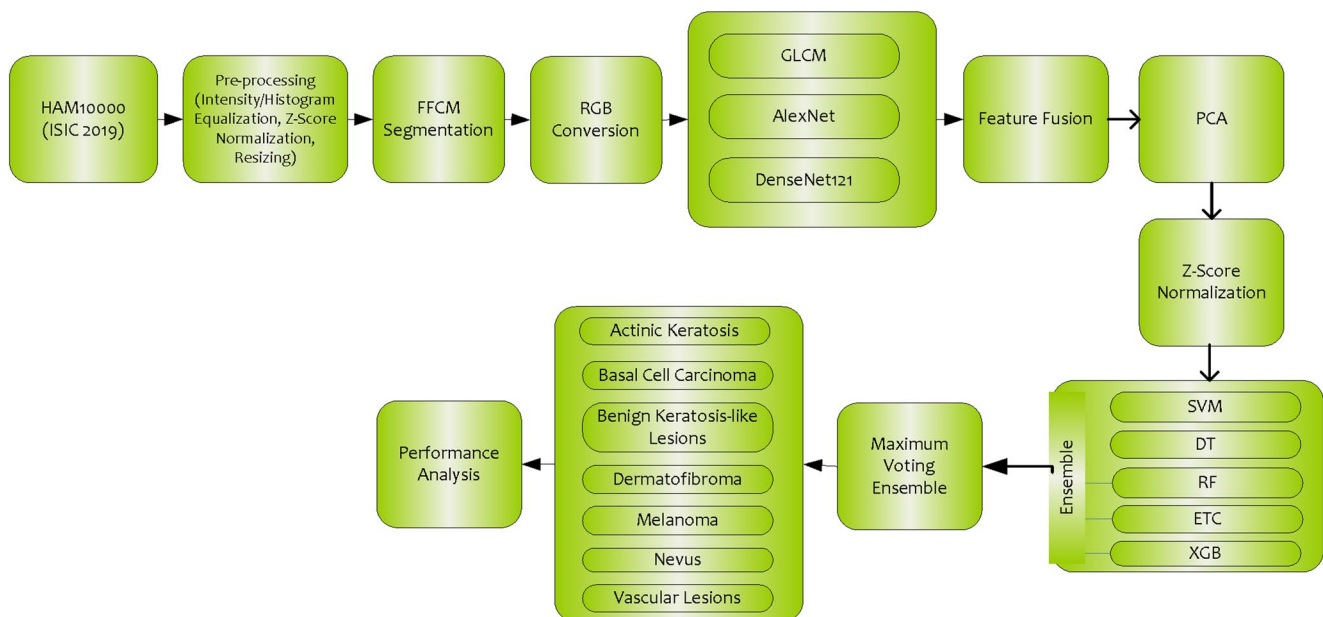


Fig. 1 Proposed SDENet skin-cancer classification model



- *Vascular Lesions (VASC).*

It typically embodies a wide-range of vascular anomalies, embodying hemangiomas and port-wine stains [92].

An illustration of the data samples is given in Fig. 2.

### B. Pre-processing.

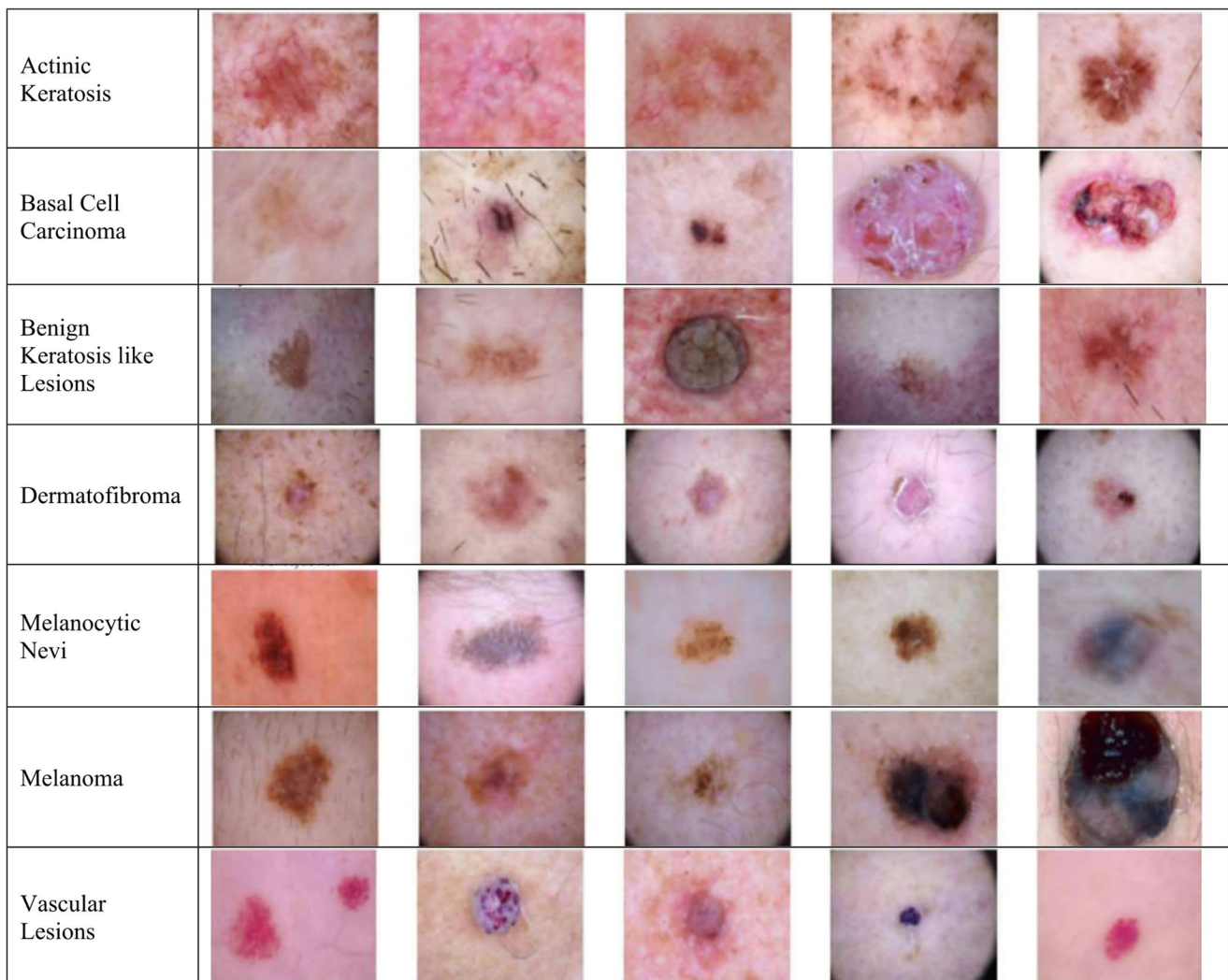
As discussed earlier, various types of skin cancer lesions exhibit substantial diversity in shape, size, color, and morphology, along with complex spatio-textural variations. Considering these differences across large and heterogeneous dermatological image datasets, appropriate pre-processing becomes essential. To create a consistent local imaging environment, several preprocessing operations were applied, including intensity normalization, histogram equalization, z-score normalization, and image resizing. In this study, a contrast-adaptive intensity equalization

technique was employed to enhance local contrast and texture details. Specifically, Eq. (1) was used to perform contrast-adaptive histogram and intensity equalization, which effectively improves local contrast features in each dermatological image and highlights subtle lesion boundaries.

$$p_n = \left( \frac{[u_w(p) - u_w(\text{Min})]}{[u_w(\text{Max}) - u_w(\text{Min})]} \right) \quad (1)$$

In (1),  $p$  refers the pixel information pertaining to the dermatological images, while  $\text{Min}$  and  $\text{Max}$  variable refers the minimum and the maximum intensity values within an input moving window.  $u_w$  refers the expected output intensity values (i.e., the minimum and maximum). The target intensity value is obtained as per (2).

$$u_w = \left[ 1 + \exp \left( \frac{f_w - p}{s_w} \right) \right]^{-1} \quad (2)$$



**Fig. 2** An illustration of the different skin cancer (dermatological) image samples

In (2),  $f_w$  and  $\sigma_w$  refers the mean and standard deviation values in each moving window. These statistical parameters can be derived by using Eqs. (3) and (4), correspondingly.

$$f_w = \frac{1}{N^2} \sum_{(i,j) \in w(x,y)} p(i,j) \quad (3)$$

$$\sigma_w = \frac{1}{N^2} \sum_{(i,j) \in w(x,y)} (p(i,j) - f_w)^2 \quad (4)$$

In (4),  $N$  refers the size of each moving window, while  $x$  and  $y$  refers the pixel's positions over the moving window. Thus, estimating the values of (2) the intensity equalization was done. Additionally, the histogram equalization was performed by applying MATLAB function (i.e., *imhist*).

Once performing above stated intensity and histogram equalization, z-score normalization was performed so as to normalize pixel's distributions. To perform z-score normalization, the mean ( $\mu$ ) and standard deviation ( $\sigma$ ) values were obtained from each dermatological image, where the focus was to maintain the mean value as zero. Similarly, it requires the value of standard deviation is maintained at unit value 1. Consider,  $i$  be the input dermatological skin lesion image, then the normalized z-score value ( $x$ ) for each input image can be estimated as per the Eq. (5).

$$x^i = \frac{(x - \mu)}{\sigma} \quad (5)$$

The normalized images are then processed for resizing, where each input image was resized to the  $320 \times 240$  pixels. Now, once the pre-processing is done on each input image, the SDENet performed FFCM-based semantic segmentation. A brief of the FFCM segmentation method is given in the subsequent section.

### C. FFCM Semantic Segmentation for Skin Lesion (ROI) Detection and ROI-Specific RGB Conversion.

In real-world dermatological images, the size, shape and allied textural or color features might vary. Moreover, the difference in morphological aspects, especially over the different kinds of the skin-lesion images makes it difficult to classify. On the other hand, the images acquired by the real-time vision sensors can acquire non-ROI elements (i.e., lesion free area) as well, where the smaller size of lesion can undergo exceedingly high minority representation, while the non-ROI regions would behave as a majority class instance or vice-versa. In this case, extracting features from the exceedingly high majority class samples (say, non-ROI) and subsequent learning can impact overall prediction results. To alleviate such instance-wise class-imbalance retaining

ROI specific areas (depicting the skin lesion regions/pixels only) can be of great significance. To achieve aforesaid ROI-specific areas, segmentation method can be applied. Though, in the past the authors have applied static threshold-based methods such as the Otsu methods and clustering techniques to segment skin lesions [58–60]; however, their efficacy over non-linear feature space or ambiguous morphologies and edges remains suspicious. Though, the use of clustering methods such as the K-Means clustering and FCM clustering [63] can be applicable towards skin lesion segmentation; however, ensuring optimality of the cluster centroid in K-Means and the membership function in FCM remains challenge. On the contrary, any inappropriate selection of centroid and membership functions or scaling factors can result inaccurate ROI segmentation that eventually can impact the optimality of the features extracted and the learning results. To alleviate such challenges and enable a robust and automatic skin-lesion segmentation method, in this work a heuristically optimized clustering algorithm is proposed. More specifically, in SDENet model, FFCM algorithm is applied for skin lesion segmentation. Noticeably, the use of Firefly heuristic algorithm serves dual tasks, first it optimizes or tunes the cost functions to achieve highly accurate ROI segmentation, and second it enables automatic ROI (semantic) segmentation to cope up scalable and run-time demands, where manual seed point definition and centroid assignment is not possible (as in case of the classical K-Means clustering based segmentation). Unlike traditional heuristic methods such as the genetic algorithm (GA), particle swarm optimization (PSO) and equivalent algorithms, which are often criticized due to the increased computational costs because of the higher generation counts and subsequent search spaces, the Firefly algorithm demands lower hyperparameters and even serves pixel's clustering to achieve accurate and swift skin-lesion segmentation. Thus, the proposed FFCM segmentation method can enable automatic and semantic segmentation of the skin lesions irrespective of their shape, size and other morphological characteristics. It can enable to retain the ROI-specific regions, which can later be considered for ROI-specific feature extraction and learning. The detailed discussion of the proposed FFCM segmentation method is given in the subsequent sections.

## Fuzzy C-Means

As stated above, in this work, FFCM is designed as an optimized version of FCM clustering method. Here, Firefly heuristic method is applied to optimize the membership function of the FCM method. In this reference, before discussing the

FFCM-based segmentation, a brief of the FCM clustering method is given in this section.

Let,  $x_i$  refers a unit pixel present within over the input dermatological image. Over then  $n$  distributed pixels, the FCM algorithm clusters the  $n$  pixels to generate  $x = \{x_1, x_2, x_3, \dots, x_n\}$ , where  $x_{i \in d}$  states the feature vector embodying  $d$  real-valued measurements depicting  $x_i$  feature for the input pattern. Unlike traditional clustering algorithms like K-Means, FCM method makes use of the fuzzy aspects to define membership functions (MF) that improves clustering accuracy over non-linear and unknown patterns or lesion conditions (say, non-linear spatio-textural patterns). The ability of FCM to map the distinct pixels to their corresponding clusters make it suitable for the skin-lesion detection and segmentation tasks. Typically, FCM algorithm can be realized in two distinct manners Fuzzy-hard and Fuzzy-soft. In case of Fuzzy-hard realization it splits input  $x$  into multiple clusters  $G_1, G_2, G_3, \dots, G_c$ . Here, a unit instance can be a part of only one cluster. Unlike Fuzzy-Hard method, in Fuzzy-Soft method, a pixel  $x$  can belong to the different or multiple clusters. In the targeted skin-lesion segmentation problem, there might be multiple tumor or granules depicting multiple skin lesions and sub-structures in a single dermatological image. Moreover, over ambiguous edges, the Fuzzy-soft FCM clustering method can result superior accuracy for the segmentation. The proposed FFCM model applied Fuzzy-Soft mechanism to perform skin-lesion segmentation. In this method, the, it measures clustering outputs as the membership matrix, called Fuzzy Partition Matrix  $U = [u_{ij}]_{(c,n)}$ . The proposed Fuzzy-Soft method measures FMF for each cluster as per (6).

$$M_{FCN} = \left\{ u \cdot R^{c,n} \left| \sum_{j=1}^c u_{ij}, 0 < \sum_{j=1}^n u_{tj} < n \right. \right\} \quad (6)$$

In (6),  $u_{ij} \in [0,1]$  represents the membership function for  $i$ -th unique pattern present over  $j$ -th cluster, preconditioned at  $U_{ij} \in [0,1]$ , when  $1 \leq j \leq c$  and  $1 \leq i \leq n$ . Here, FCM was used over input segmented skin lesion and minimizes the objective function for clustering (6).

$$J_m = \sum_{j=1}^c \sum_{i=1}^n u_{ij}^m \|x_i - v_j\|^2 \quad (7)$$

In (7), states the centroid of the skin lesion(s), which is measured as the inner-product from to the centroid of the  $h$  cluster. In this work, FMF signifying the level of fuzziness to the pattern outputs by applying weight parameter. Functionally, it follows the subsequent steps to perform clustering.

In (7),  $\{v_j\}_{j=1}^c$  states the centroid of the skin lesion(s), which is measured as the inner-product from  $x_i$  to

the centroid of the  $j$ -th cluster. In this work, FMF signifying the level of fuzziness to the pattern outputs by applying weight parameter  $m \in [1, \infty]$ . Functionally, it follows the subsequent steps to perform clustering.

- Choose  $C$  centroid arbitrarily and initiate clustering.
- Measure the Fuzzy membership function for each pixel available in each cluster.
- Measure the centroid iteratively by reducing objective function.

Apply (8) to measure membership value and tune the centroid until there is no change in the (centroid) value.

$$u_{ij} = \frac{1}{\sum_{k=1}^c \left( \frac{\|x_i - v_j\|}{\|x_i - v_k\|} \right)^{\frac{2}{m-1}}} \quad (8)$$

In traditional FCM clustering method, the centroid is calculated and updated as per the Eq. (9). Yet, its efficacy remains unclear over ambiguous, non-linear dermoscopic images.

$$v_j = \frac{\sum_{i=1}^n u_{ij}^m \cdot x_i}{\sum_{i=1}^n u_{ij}^m} \quad (9)$$

To address aforesaid challenge and enable accurate ROI (i.e., skin lesion) segmentation over input ROI-specific skin lesion, which are typically non-linear and possess unclear edge, this research proposes Firefly heuristic-based FCM (FFCM). Here, Firefly algorithm mainly focuses on optimizing centroid values and corresponding automatic clustering (say, instance mapping) to achieve self-driven ROI segmentation. A brief of the FFCM-based segmentation method is given in the subsequent sections.

## 1. FFCM-based Skin-Lesion detection and segmentation

To optimize clustering accuracy the Firefly heuristic method minimizes inter-cluster distance which helps to enhance centroid of each cluster to refine clustering output. Functionally, it is designed on the basis of the Firefly's light generation pattern and subsequent pixel's movement characteristics. It makes use of the light pattern (say, Firefly's lighting behavior) as a signal to attract adjoining fireflies. Hypothesizing each pixel as Firefly, it influences neighboring Fireflies (say, pixels) to the similar cluster and it improves clustering-driven segmentation accuracy. For automatic ROI segmentation, FFCM applies certain rules, which are given as follows:

*Rule-1: Fireflies being unisexual, it can be attracted to each other on the basis of the relative light behavior, which are independent of any gender factor.*

*Rule-2: Fireflies typically influence other Firefly in reference to their corresponding brightness; though, they attract other Fireflies inversely to the distance. A Firefly possessing higher illumination attracts other Fireflies with relatively lower brightness. A Firefly with the higher inter-instance distance influences other Firefly less in comparison to the one having higher brightness. In this manner, it follows the condition  $I \propto \frac{1}{r^2}$ , where  $r^2$  being the square of the distance. Higher value of  $r^2$  results reduced coefficient of attraction  $\beta$ . Thus, a Firefly with the highest brightness and with no neighboring Firefly traverses randomly in the direction in which the Firefly has the higher brightness.*

*Rule-3: A firefly possessing the maximum illumination can remain moving arbitrarily across the search space.*

The FFCM initially estimates the optimal set of centroids for each dermatological image (say, ROI specific skin lesion) to segment skin lesion(s). It at first, estimates the centroid value in  $N$ -dimensional search space by using correlation between the cost-function and the inter-instance distance (i.e., the distance between pixels). At first, it distributes Fireflies arbitrarily over the search space. Here, it measures the intensity differences, which is followed by the pixel's movement to constitute cluster (i.e., skin lesion). The difference of intensities is applied as a cost function, where the Firefly with lower or higher brightness attracts other Firefly with the relatively higher or lower brightness, correspondingly. It enables pixel's clustering in such manner that the pixels with the similar patterns are clustered together. Let, there be  $n$  Fireflies in the search-space and  $x_i$  be the solution representing the  $i$ -th Firefly. Then,  $f(x_i)$  refers the fitness value for  $x$ , with the brightness value  $I$ . Here,  $f(x_i)$  measures the real pixel's position. The brightness value is measured for each Firefly by applying Eq. (10).

$$I_i = f(x_i) \quad 1 \leq i \leq n \quad (10)$$

Estimating the Firefly's brightness, the proposed method initiates swarm movement where each Firefly traverses to the other Fireflies based on the relative brightness values. As stated earlier, the attractiveness of a Firefly is always directly proportional to the brightness of the other Fireflies. Typically, each Firefly possesses a definite attractiveness  $\beta$  which primarily depends on the distance  $r_{ij}$  between the  $i$ -th and  $j$ -th Firefly, present across the search space.

$$r_{ij} = |x_i - x_j| \quad (11)$$

$$\beta(r) = \beta_0 \exp\{-\gamma d(i,j)^2\} \quad (12)$$

In (11),  $\beta_0$  states the attractiveness, when the Fireflies are present at the same location (i.e.,  $r = 0$ ). The parameter  $\gamma$  signifies the light absorption-coefficient, hypothesizing that the Firefly  $j$  has higher brightness than the  $i$ -th Firefly. Thus, the Firefly  $i$  traverses towards the  $j$  Firefly, where the traversed distance can be measured (13). In (12),  $rand$  states a random number generator yielding arbitrary number existing in between 0 and 1.

$$x_i(t+1) = x_i(x_j - x_i) + a(rand - 0.5) \quad (13)$$

The deployed Firefly algorithm calculates the optimal location or the best position of a Firefly, signifying the centroid for a cluster or the pattern, and thus retrieves an array of solutions  $A$  (14). In (14),  $a_i$  represents the centroid while  $a_i \in A$ . In (14),  $s_i$  represents the centroid in reference to the feature elements  $d\{a_1, a_2, \dots, a_d\}$ . The centroid values are and updated to the FCM to cluster skin lesion (s) accurately. The mechanism of centroid tuning continues until the values turn out to be constant, and thus the final centroid value is considered to be the eventual, with respect to which the clustering is performed.

$$A = (s_1\{a_1, a_2, \dots, a_d\}, s_2\{a_1, a_2, \dots, a_d\}, s_3\{a_1, a_2, \dots, a_d\}) \quad (14)$$

In this work, a total of 30 Fireflies were deployed (as initial population) ( $n=25$ ). The number of iterations (here, 100) was considered as the stopping criteria, while brightness coefficient was maintained at  $\beta = 1$  and  $g = 1$ . Here, Rosenbrock valley function was considered as the cost function so as to enhance spatial clustering accuracy in dermatological image input. Additionally, it enables swift convergence in comparison to the other cost-functions. Noticeably, the fitness score represents the extent to which a pixel can become a good or bad centroid. A Firefly possessing higher brightness is stated to have lower fitness and therefore it influences other Fireflies within the search space. In this manner, it controls the movement of the pixels within the search space for accurate clustering performance. Consider,  $a$  and  $b$  represents the Fireflies, where  $b$  possesses higher brightness than  $a$ . Then, the Firefly  $a$  traverses towards  $b$  Firefly. It continues over the iterations till it approaches the stopping criteria. The FFCM method retrieves the tuned solution  $a' = a'_1, a'_2, a'_{31}, \dots, a'_N$ , by using  $f(a')$ . The FFCM method applies Rosenbrock valley function, where the global minima occur over a narrow, long and parabolic shaped flat valley. Despite, identifying the valley region a challenging task, the solutions (14) can't be assumed to have ('1') value (15).

$$f(x, y) = (a - x)^2 + b(y - x^2)^2 \quad (15)$$



To achieve global minima at pixel's position  $(x, y) = (a, a^2)$ , it requires fulfilling  $f(x, y) = 0$ , which is defined in such way that it generates  $a = 1$  and  $b = 100$ , and therefore with  $a = 0$ , it yields a symmetric function having the minima at the origin, for which the Eq. (16) was applied.

$$f(x) = \sum_{i=1}^{N-1} 100(x_{i+1} - x_i^2)^2(1 - x_i)^2 \quad (16)$$

In (16), carries unit minima having. Despite it contains two minima, conditioned at it enables achieving the expected global minima having all 1s, primarily when the local minima exist near. The proposed FFCM model used this condition by assigning the gradient of the function as zero, as the function stated in (16) being rational function of. Thus, applying this method, the proposed FFCM segmentation model segmented each ROIs present over input ROI-specific images. Once segmenting the skin lesions, the segmented areas were multiplied with the original RGB images that eventually generated the ROI-specific color representation, which was passed as input for the proposed feature extraction model.

In (16),  $X = [x_1, x_2, \dots, x_N] \in \mathbb{R}^N$  carries unit minima having  $N = 3$  (at (1,1,1)). Despite it contains two minima, conditioned at  $4 \leq N \leq 7$  it enables achieving the expected global minima having all 1s, primarily when the local minima exist near  $(x_1, x_2, \dots, x_N) = (-1, 1, \dots, 1)$ . The proposed FFCM model used this condition by assigning the gradient of the function as zero, as the function stated in (16) being rational function of  $x$ . Thus, applying this method, the proposed FFCM segmentation model segmented each ROIs present over input ROI-specific images. Once segmenting the skin lesions, the segmented areas were multiplied with the original RGB images that eventually generated the ROI-specific color representation, which was passed as input for the proposed feature extraction model.

Unlike traditional skin lesion classification systems, in this work multiple feature extraction modules were applied in parallel to extract deep spatio-textural features, including the GLCM, DenseNet121 and AlexNet. Each of these feature models yield distinct feature space for further fusion and learning. The detailed discussion of the proposed feature extraction model is given in the subsequent sections.

The accuracy of lesion segmentation plays a crucial role in ensuring the quality of subsequent feature learning. The Firefly-optimized Fuzzy C-Means (FFCM) algorithm not only isolates the region of interest but also minimizes background noise and irrelevant pixels that could otherwise bias feature extraction. By enhancing the structural and textural clarity of the segmented lesion, the model provides more

reliable inputs to the **DeS-TEFF** module, enabling the deep networks (AlexNet and DenseNet121) to focus on diagnostically relevant patterns rather than background artifacts [93].

#### D. ROI-Specific Deep Spatio-Textural Feature Ensemble and Fusion (DeS-TEFF).

In this work, the focus is to extract sufficiently large feature space, which could be inputted to the learning model to yield accurate skin-cancer detection and classification. Recalling the fact that the traditional CNN methods or RNNs such as the LSTM deep networks mainly exploit the local hierarchical features and lack contextual details, in this work the focus is made on applying DenseNet121 that being a residual network model can serve high-dimensional residual features for learning and prediction. On the other hand, it also applies AlexNet model that can serve 4096-dimensional features to achieve higher learning efficiency.

The choice of combining AlexNet and DenseNet121 was driven by their complementary feature-learning characteristics. AlexNet, being relatively shallow with larger convolutional kernels, effectively captures low- to mid-level texture and color variations across lesion regions. In contrast, DenseNet121, with its deeper structure and dense connectivity, extracts high-level semantic and contextual representations through feature reuse. Integrating their feature maps enables the model to leverage both global lesion structure and fine-grained local details. To minimize feature redundancy, Principal Component Analysis (PCA) was applied to the concatenated feature vector, retaining only the most discriminative components for ensemble classification. This fusion strategy enhances feature diversity and strengthens classification robustness against inter-class similarities.

In addition, the use of GLCM features, especially pertaining to the textural, orientation and statistical distribution can provide feature diversity with the DenseNet121 and AlexNet to achieve improved learning and prediction solution. The features extracted from these models are then fused to yield a composite feature vector which has been passed as input to the E2E ensemble framework to perform learning and prediction. The detailed discussion of the proposed deep spatio-textural feature extraction and fusion model is given in the subsequent sections.

## 1. GLCM Spatio-Textural feature extraction

The GLCM being a statistical descriptive feature model can yield a large number of spatio-textural, orientational and descriptive features from the input images. For instance, it extracts mean, standard deviation, variance, etc. as the

statistical distribution features, while energy, entropy, correlation, homogeneity etc. can be obtained as the spatio-textural features. On the other hand, it generates kurtosis and skewness as the orientational features. These features cumulatively constitute a large set of features for learning and prediction tasks. Its efficiency can further be improved by amalgamating deep features to enable more efficient learning and prediction. In this work, the ROI-specific color images (i.e., skin lesions) are fed as input to the GLCM to extract the different spatio-textural features including energy, entropy, mean, standard deviation, correlation, homogeneity, skewness, kurtosis. A brief of these features is given as follows:

To extract GLCM features, for each input skin lesion the features are extracted in the form of a matrix representing pixel intensities  $I(x, y)$ . Obtaining the pixel's intensities, the probability matrix  $P_{i,j}$  is measured for each ROI-specific skin lesion. The disparity of intensities between  $i$ -th and  $j$ -th pixel enables detection of the motion pattern(s). Yet, the gray-scale value refers the pair-association in one direction. With the obtained gray-information, it measures corresponding transpose matrix. Additionally, the gray-scale information is applied to derive the symmetric matrix  $S$  that with its corresponding transpose matrix measures the associations in a direction. The proposed method made use of the Eq. (17) to measure the probability matrix  $P$ .

$$P_{i,j} = \frac{S_{i,j}}{\sum_{i,j=0}^{N-1} S_{i,j}} \quad (17)$$

- *Contrast.*

It signifies the alterations in the gray scale values over input skin-lesion images. Typically, the pixel pairs present over the diagonal elements of  $P_{i,j}$  (17) refers the high-contrast difference. One can also infer it as the higher change in the gray scale values across the diagonal matrix of (17) indicates that the image has higher (change in) contrast values. It can also be stated as the overall change in the pixel intensities across the input skin lesions.

- *Homogeneity.*

For GLCM feature extraction, homogeneity signifies the Inverse Different Moment (IDM), depicting that the lower contrast value can have the higher homogeneity value. In this work, the Eq. (18) was applied to measure the homogeneity across the skin lesion images.

$$HOM = \sum_{i,j=0}^{N-1} \frac{P_{i,j}}{1 + (i - j)^2} \quad (18)$$

The traditional GLCM features used to have linear or the uniformly distributed features across the grids. In case, the GLCM possesses small component of the major magnitude, it indicates higher contrast and hence low homogeneity. The contrast is measured as per the Eq. (19).

$$CONT = \sum_{i,j=0}^{N-1} P_{i,j} (i - j)^2 \quad (19)$$

- *Energy.*

To measure energy of each skin lesion (image), at first angular second moment (ASM) information was measured. Typically, ASM assesses the rotational acceleration throughout the segmented skin lesions, and is measured as per (20). The value of ASN increases with increase in linear distribution of the gray-scale values. The energy feature (21) is measured by using ASM value, as depicted in (20).

$$ASM = \sum_{i,j=0}^{N-1} P_{i,j}^2 \quad (20)$$

$$ENR = \sqrt{ASM_{i,j}} \quad (21)$$

- *Entropy.*

It refers the extent to which the gray-scale values are non-linearly distributed across the skin lesion areas. With increasing non-linearity in pixel values, the entropy increases pixel's non-linearly (distribution). This work applied Eq. (22) to calculate entropy feature of each skin lesion image.

$$ENT = \sum_{i,j=0}^{N-1} P_{i,j} (-\ln P_{i,j}) \quad (22)$$

- *Correlation.*

As statistical features, four features named mean, standard deviation, variance and correlation were obtained. These statistical features represent the statistical characteristics pertaining to the pixel's pair (gray scale) relationship, which can differ from the gray scale values. Therefore, the inter-pixel relationship defines aforesaid descriptive features depicting unique pattern to help making classification decision. In this work, Eq. (23) was applied to measure mean of the two distinct pixels. Because of the symmetric in nature (17), the mean-values (23) remain same.

$$\mu_j = \sum_{i,j}^{N-1} j (P_{i,j}) \quad (23)$$

To have high feature heterogeneity, standard deviation (25) and variance (24) were obtained.

$$\begin{aligned}\sigma_i^2 &= \sum_{i,j}^{N-1} P_{i,j} (i - \mu_i)^2 \\ \sigma_j^2 &= \sum_{i,j}^{N-1} P_{i,j} (j - \mu_j)^2\end{aligned}\quad (24)$$

$$\begin{aligned}\sigma_i &= \sqrt{\sigma_i^2} \\ \sigma_j &= \sqrt{\sigma_j^2}\end{aligned}\quad (25)$$

Exploiting above derived mean and standard deviation and variance information, the correlation was measured by using (26).

$$CORR = \sum_{i,j}^{N-1} P_{i,j} \left[ \frac{(i - \mu_i)(j - \mu_j)}{\sqrt{(\sigma_i^2)(\sigma_j^2)}} \right] \quad (26)$$

Thus, extracting above features, horizontal concatenation is performed to obtain the composite spatio-textural feature (27).

$$\begin{aligned}Feat_{GLCM} &= [CONT, ENE, ENT, HOMO, CORR, VAR, Mean, SD] \\ &= [CONT, ENE, ENT, HOMO, CORR, VAR, Mean, SD]\end{aligned}\quad (27)$$

## 2. AlexNet- driven feature extraction

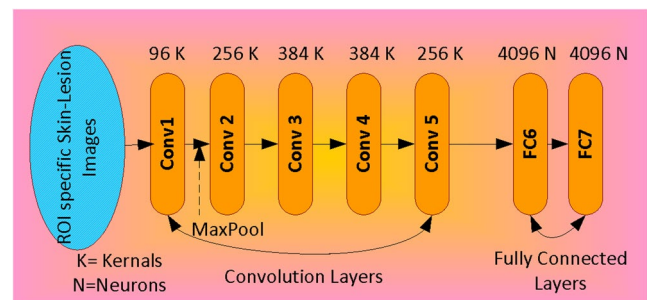
This is the matter of act that a majority of the existing methods have applied traditional CNN, DNN and RNN variants to perform skin lesion classification; however, majority of the state-of-arts forget to address certain challenging issues such as the lack of contextual details, gradient vanishing, gradient explode and long-term dependency. The CNN models primarily exploit local hierarchical features to perform skin cancer classification. The lack of contextual details can limit their efficacy over the ambiguous edges and non-linear morphologies. Similarly, the DNN models, specially designed with the larger convolutional layers undergo gradient vanishing and gradient explode problem, which can have the direct impact on learning and prediction. To address such challenges, this work applies DenseNet121 deep network along with the AlexNet, where the earlier can provide finer grained and information rich feature, while the later helps to serve high-dimensional feature are given in Table 1.

A brief of these deep learning models is given in the subsequent sections:

A typical structure of the AlexNet deep network is given in Fig. 3. As depicted the AlexNet deep network embodies five convolutional layers (CONV) and three fully connected layer (FC). The ability to use the different kernel sizes over increasing convolutional layers make AlexNet capable to retrieve high-dimensional information rich, high-resolution features. On the other hand, the high-dimensional FC layer

**Table 1** Deep feature models and their significance

| Model       | Features                                                   | Significance                                                                                                                          |
|-------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| DenseNet121 | Feature Reuse and Layer-wise learning<br>Residual Learning | Reduced Hyperparameters and hence reduced computational costs<br>Avoids gradient vanishing, gradient explode and accuracy degradation |
| AlexNet     | High-dimensional (4096) features<br>Lightweight in Design  | Reduced computational costs<br>Dynamic kernels over subsequent convolutional layers retain high-resolution multi-attention features   |



**Fig. 3** AlexNet deep network

(4096-dimensional) yields sufficiently large feature space to train the model to yield accurate prediction results.

As illustrated in Fig. 3, AlexNet deep model embodies five CONV layers, two FC layers, where each of the convolutional layer is designed with the different kernel sizes so as to retrieve high-resolution features. CONV1 is designed with the kernel size of 96, while CONV3 and CONV4 were assigned the kernel sizes of 384 each. The CONV5 layer was designed with 256 kernels. Here, after each convolutional layer, unit Max-pooling layer was applied to retain high-resolution features. In addition, drop-out layer with the filter coefficient of 0.5 was used to retain the optimal set of features, while dropping the redundant feature elements. The convolutional feature was then mapped as input to the FC layer where FC6 and FC7 yields 4096 dimensional features, while FC8 provides 1024-dimensional feature. In this work, FC6 feature was considered for further fusion and learning. The design specifications for the proposed AlexNet deep network are given in Table 2.

The ROI specific skin lesion images were inputted to the AlexNet input layer, where the images were resized to the dimension of  $320 \times 240$ . At the input layer, the linear activation function was considered and therefore it yielded the same output as input skin lesion image. The output of the input layer was then passed as input to the CONV layer that applied different convolutional filters to extract spatial features. More specifically, the proposed method applied horizontal and vertical filters to generate local features,

which was later processed for max-pooling layer, which retained high-resolution features. The outputs of the first convolutional layer (CONV1), which applied 96-kernel for feature extraction was passed as input to the CONV2 which retrieved 256-dimensional features, followed by 384-dimensional feature extraction by CONV3. As depicted in Fig. 3, CONV5 extracted 256 dimensional features. During the feature extraction, each neuron extracted the feature map that shares similar weights ( $W$ ) and bias ( $b$ ). The similar values of  $w$  and  $b$  makes neurons identifying same pattern over input images and its features. The feature maps make use of the distinct sets of  $w$  and  $b$  values to extract the local features. In this work, ReLU layer was applied as the linear activation function. Here, single ReLU unit was applied with each CONV layer. Consider,  $x$  represents the input then the applied ReLU layer assessed neuron's output  $q(x)$  as  $x$  pre-conditioned at  $x > 0$ . Now,  $x \leq 0$ , ReLU generated  $(\delta \times x)$  as output, where  $\delta$  signifies a condition whether the non-positive elements can be avoided by multiplying it with a slope value 0.01 or 0.1 or can it be fixed at zero. Here,  $\delta$  was assigned as 0, which forced ReLU to act as a default layer in reference to the function  $q(x) = \max(0, x)$ . Here, the Max-pooling layer is applied as a consolidation layer which retains high-resolution features for learning. In fact, Max-pooling layer acts as a feature selection later which reduces the spatial resolution of each feature map retrieved by each CONV layer. In this manner, it reduces the total hyperparameters required and hence improves computational efficiency and cost. Functionally, the deployed deep model performed averaging followed by sub-sampling, which alleviated the over-fitting issues. Moreover, it helped to retain translation-invariant features by down-sampling the latent feature representation. In the deployed model, AlexNet model exhibited feature scaling followed by averaging subtraction. In this work, AlexNet was applied as a feature extraction model, where the features obtained were used as aggregated feature for further learning and prediction. The FC layer feature was

used as a classification layer by performing high-level reasoning. We used FC6 layer functions as a concatenation layer that in conjunction with the averaging pooling generated composite deep feature vector for further learning and prediction. The FC layer concatenated the CONV features obtained from the Max-pooling layer retrieved through the different neurons, which was subsequently mapped to the connected neurons to output 1-D high-resolution features. The extracted features were then fused with the other features (i.e., GLCM and DenseNet121) to yield composite feature, which is later used to perform learning towards skin-cancer classification.

### 3. DenseNet121 feature extraction

Unlike traditional CNN or DNN architectures, DenseNet121 deep network extracts deep residual features for further learning and prediction, where the ability to perform feature reuse and layer-wise learning yields information rich feature to perform learning and prediction. Here, DenseNet makes use of the classical CNN (ConvNet) architecture to extract feature, which is then amalgamated with the different filters generates a high dimensional feature set. In DenseNet121 deep network, each layer gets supplementary input from the layers before it, which is subsequently traversed to a feature map in the subsequent layer. Figure 3 illustrates the DenseNet121 feature extraction. As per the standard definitions, it is a deep model with densely interfaced CONV networks where features are mapped from one layer to the next on the basis of their extraction. DenseNet121 deep model reduces the likelihood of gradient vanishing issues that improves feature reuse and thus reduces hyperparameters to enable cost-effective learning and prediction.

The ability of feature propagation over the next layers and feature reuse makes learning more efficient towards skin-cancer classification. Moreover, the mechanism of feature reuse reduces the hyperparameters and hence improves computational cost and delay. These features make DenseNet121 suitable towards the targeted skin-cancer detection and classification. Architecturally, DenseNet121 deep model embodies the Dense Blocks (Fig. 4), constituting the multiple convolution layers (Figs. 4 and 5). The FC layers which are designed as the flatten layer embodying 1000 units were interfaced to the 121 CONV layers of the DenseNet121. In other words, the DenseNet121 deep model comprises 121 CONV layers, which are interfaced to either an output FC layer. Since, in this work DenseNet121 deep network is applied as feature extraction model, the Softmax layer was substituted to the FC layer, and the extracted feature was later fused with other feature sets (i.e., GLCM and AlexNet) to perform learning and prediction.

**Table 2** Design specifications of the AlexNet deep network

| Layer Name    | Specification                                                      |
|---------------|--------------------------------------------------------------------|
| Input Layer   | $320 \times 240$ dimensional images with zero-center normalization |
| CONV          | $320 \times 240$ CONVs with three stride and zero-padding.         |
| ReLU          | Native ReLU activation layer.                                      |
| Max-Pooling   | Max-Pooling with the stride of 4 and zero-padding.                 |
| FC            | FC6 (4096 dimensional features)                                    |
| Drop-out      | 0.5 filter coefficient                                             |
| Learning Rate | 0.0001                                                             |
| Optimizer     | ADAM                                                               |
| Epoch         | 100                                                                |
| Iterations    | 200                                                                |



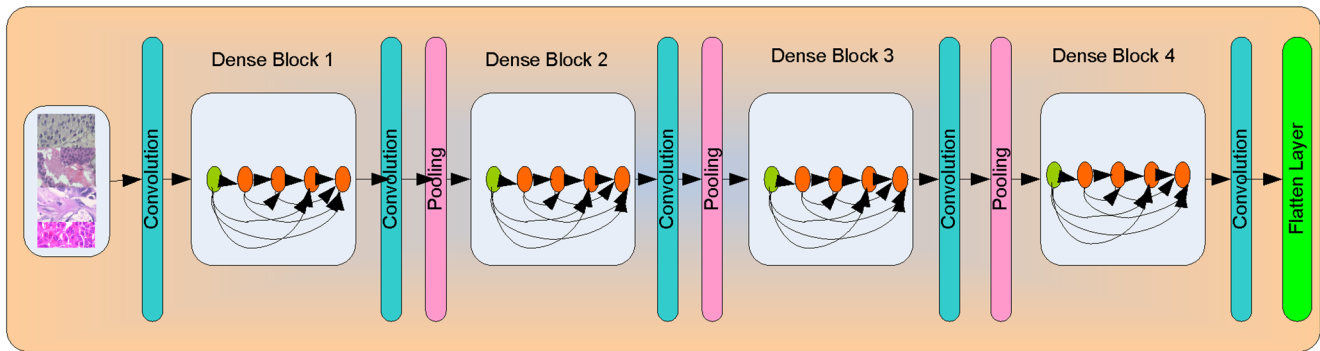
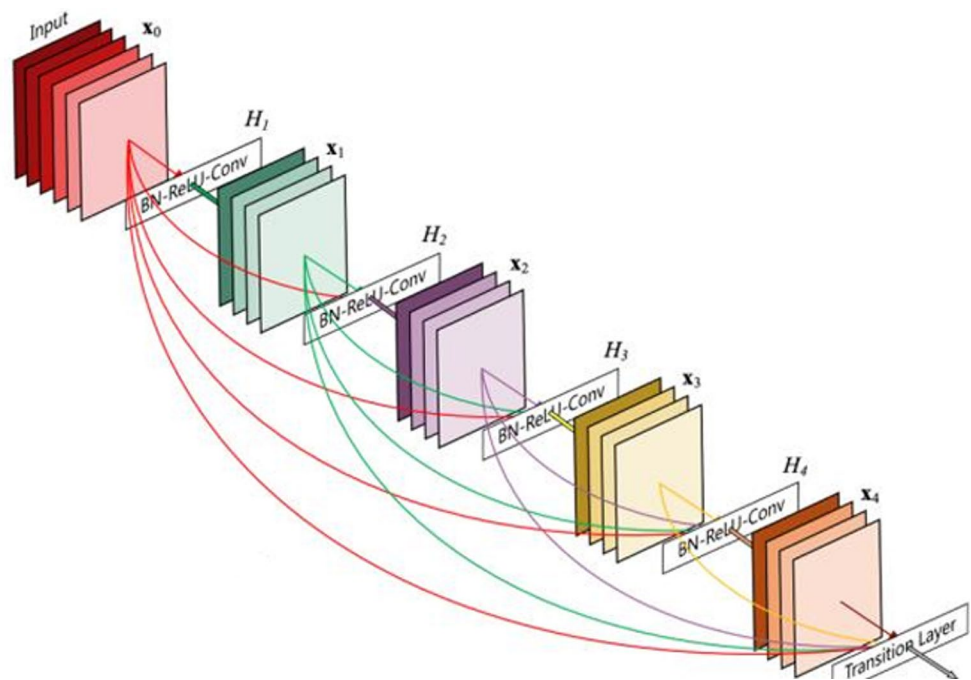


Fig. 4 DenseNet121 deep model

Fig. 5 DenseNet121 convolution and transition layer arrangement



As depicted in Figs. 4 and 5, the hierarchical features were obtained and therefore the feature maps of  $l-1$  layer is inputted to the  $l$  layer. The feed-forward network interfaces the output of the  $l$  layer to the subsequent  $(l+1)$  layer, and thus served source to the next layer transition (i.e.,  $X_1 = H_1(X_1 - 1)$ , where  $l$  represents the layer's index.  $H_1$  states the composite function embodying functions like batch normalization, ReLU, Max-pooling layers etc. The feature maps pertaining to the earlier layer in deep network are fused and feed as input to the specific layer (Fig. 4). Unlike traditional residual networks, where the output of one layer is connected to the very subsequent layer, the output from the DenseNet121 deep model is mapped to the specific layer that subsequently reduces the hyperparameters and hence computational costs. This method minimizes feature space and hence suppresses redundant computation to meet run-time CAD demands. It obtained the optimal set of features by mapping features to the subsequent layer for

learning, thereby alleviating the gradient vanishing problems. Once combining the feature maps from the earlier layers ( $X_0, X_1, \dots, X_{l-1}$ ) and inputting to the  $X$  layer, the  $l$  layer retrieved the feature maps from the earlier layers (28).

$$X_l = H_1([X_0, X_1, \dots, X_{l-1}]) \quad (28)$$

In (28),  $[X_0, X_1, \dots, X_{l-1}]$  states the combined feature map obtained over the layers  $0, 1, \dots, l-1$ . In the proposed DenseNet121 network, the inputs of  $H_1()$  were concatenated to yield a single tensor to generate composite dense feature for learning and prediction.

As depicted in Fig. 4, the DenseNet121 deep model amalgamated four Dense Blocks (DB) with one unit transition layer connected between each DB (Fig. 4). Each DB encompasses a CONV layer, and the transition layer amalgamates features such as the batch normalization, Max-pooling layer, activation function and clipped ReLU (cReLU). Here,

cReLU layer was applied as the activation function, where a value below the clipping ceiling is defined as zero, and a value more than the clipping ceiling threshold is defined to that ceiling. The design specifications and connections amongst the different layers are shown in Fig. 5. The DenseNet121 uses cReLU activation function to improve feature non-linearity learning, making it more sensitive and information rich. The feature extracted at the last layer was processed by the global average pooling layer, where the extracted features  $Feat_{DenseNet121}$  were later fused to yield a composite feature set for learning and classification.

Once extracting the deep (AlexNet  $Feat_{AlexNet}$  and DenseNet121  $Feat_{DenseNet121}$ ) and GLCM features ( $Feat_{GLCM}$ ), feature level fusion was performed to obtain the composite feature set ( $Feat_{DeS-TEFF}$ ) for learning and prediction. More specifically, in this work robust horizontal concatenation was performed to generate the composite feature vector (29).

$$Feat_{DeS-TEFF} = [Feat_{GLCM}; Feat_{AlexNet}; Feat_{DenseNet121}] \quad (29)$$

Now, once obtaining the composite feature set  $Feat_{DeS-TEFF}$ .

, the proposed method applied PCA feature selection so as to retain the most significant features while dropping the redundant one. A brief of the feature selection method applied is given in the subsequent section.

#### E. PCA Feature Selection.

This is the matter of fact that the proposed deep spatio-textural feature fusion and allied  $Feat_{DeS-TEFF}$  feature set can provide diverse and information rich feature vector for learning and prediction; however, at the cost of increased computational cost. Moreover, the increased search space in  $Feat_{DeS-TEFF}$  can give rise to the local minima and convergence issues. To alleviate this challenge, in this work feature selection is performed. To achieve it, in this work PCA feature selection method was applied. The proposed method at first measured eigen values and principal components for each feature set ( $Feat_{DeS-TEFF}$ ). In reference to the estimated principal component average value, which was defined as 0.5, the Eigen distance of every feature was measured. Thus, those features with their distance less than 0.5 were retained. On the contrary, the features (instances) with the distance  $< 0.5$  were discarded or dropped. In other words, the feature (instance) with the low Eigen values was retained for further learning and classification. It reduces the search space that eventually alleviates the probability of local minima and convergence.

#### F. Z-Score Normalization.

The selected features were processed for z-score normalization to reduce the likelihood of over-fitting. To achieve it, the selected feature sets were processed to measure the mean ( $\mu$ ) and standard deviation ( $\sigma$ ) values, where to ensure normalized feature distributions, the value of  $\sigma$  was maintained at one, while mean was intended to be maintained at zero. The z-score value ( $x$ ) of each input feature set (say, features pertaining to the skin lesion image  $i$ ) was measured as per (30).

$$x^i = \frac{(x - \mu)}{\sigma} \quad (30)$$

The normalized features were mapped as input to the proposed ensemble-of-ensemble (E2E) learning model.

#### G. Ensemble Learning based DR Severity and Grading.

Unlike traditional skin lesion classification models, where the authors have either applied cross-entropy cost function with the Softmax layer to perform classification or have used standalone machine learning algorithm such as the SVM, DT, ANN, RF etc., in this work a robust E2E ensemble model was designed based on the principle of maximum voting ensemble or consensus-based ensemble. Here, each of the base classifiers were applied distinctly to learn the features, while their corresponding prediction results were applied to make maximum-voting based prediction. In this work, three ensemble learning models, as given below were applied to constitute maximum voting -based ensemble learning and classification.

- *Support Vector Machine.*
- *Decision Tree.*
- *Random Forest (RF).*
- *Extra Tree Classifier (ETC), and.*
- *XGBoost Classifier.*

A brief of these ensemble learning models is given in the subsequent sections.

## 1. Support vector machine

SVM is a well-known machine learning model often used for text and image classification tasks, where its ability to learn input features by using hyper-planes makes it easy to implement and execute. Based on the input features, it iteratively minimizes the generalization error. To generate a training subset depicting the boundary conditions, it estimates the support vector, which is measured to assess the

hyperplane existing between the classes. This work applied Eq. (31) to perform multi-class classification.

$$Y' = w^* \varphi(x) + b \quad (31)$$

In (31),  $\varphi(x)$  signifies a non-linear transform function that assigns appropriate values of the weights  $w$  and biases  $b$  for the learning and (skin-cancer) classification. The result  $Y'$  is measured by reducing a regression-risk parameter (32).

$$R_{reg}(Y') = C \sum_{i=0}^l (Y'_i - Y_i) + \frac{1}{2} w^2 \quad (32)$$

In (32), the parameters  $C$  and  $\gamma$  refers the penalty factor and the cost-function, correspondingly. The value of weight  $w$  is measured as per (33), which is updated to the Eqs. (31) and (32) to perform learning and classification.

$$w = \sum_{j=1}^l (a_j - a_j^*) \varphi(x_j) \quad (33)$$

In Eq. (33),  $\alpha$  and  $\alpha^*$  parameters signify non-zero value, also referred as the Lagrange relaxation. In this manner, the SVM classifier annotates each input skin cancer image as the most probable class by applying function (34).

$$\begin{aligned} Y' &= \sum_{j=1}^l (a_j - a_j^*) \varphi(x_j) + b \\ &= \sum_{j=1}^l (a_j - a_j^*) K(x_j, x) + b \end{aligned} \quad (34)$$

In (34),  $K(x_j, x)$  signifies the kernel function, which was selected as the radial basis function for multi-class classification.

## 2. Decision Tree.

The decision tree (DT) algorithm is one of the most used association rule mining methods used for image classification and text analysis tasks. In function, originating from the root node, it divides the input data into multiple branches by applying association rule and certain defined split condition. At each tree node, the different branches are generated. In this work, to perform learning and prediction, the information gain ratio (IGR) was considered throughout the branches. Dividing or splitting the composite feature  $Feat_{DeS-TEFF}$  into multiple branches, it generates supplementary nodes that eventually branch off to generate additional data. It looks like a tree possessing multiple branches. It resembles a binary tree possessing a unit parent node having the nodes of multiple children, like the left and right nodes. Consider that  $LC_d$  and  $RC_d$  be the left and

right child nodes and  $I$  be the noise value and  $x$  be the input feature. Then, as per the formulation (35), the data present in  $P_d$ ,  $LC_d$  and  $RC_d$  intends to maximize information gain.

$$InformationGain(P_d x) = I(P_d) - \frac{LC_n}{P_n} I(L.C_d) - \frac{RC_n}{P_n} I(R.C_d) \quad (35)$$

In Eq. (35),  $I$  is measured by making use of the Gini-Index  $I_G$  (37), entropy  $I_H$  (36) or the classification error  $I_E$  (38).

$$I_H(n) = - \sum_{i=1}^c p(c|n) \log_2 p(c|n) \quad (36)$$

$$I_G(n) = 1 - \sum_{i=1}^c p(c|n)^2 \quad (37)$$

$$I_E(n) = 1 - \max \{p(c|n)\} \quad (38)$$

In (36-38), the parameters  $c$  and  $n$  signifies the class(es) and corresponding node  $n$ . In this work, the probability parameter is obtained in the form of a ratio of  $c$  to  $n$ . In this manner, learning over each input sample it classifies input skin lesion images into the multiple categories and assigns corresponding labels, which is later applied by MVE to make final prediction.

## 3. Random forest

Typically, the maximum voting-based decisions are made by applying multiple base classifiers, where each classifier learns and yields predicted results to make eventual prediction decisions. In RF ensemble method, the voting-driven learning and prediction mechanism uses multiple tree-based (base) classifiers, where each tree learning model provides or yields unit vote to each skin-lesion image based on its most probable class (i.e., the likelihood of the belongingness). The highest voting score or value for each skin lesion image is measured by deploying a definite number of trees over the input feature. Thus, an image with the maximum score is used to make final decision. Consider that there are  $NNMmh\{\theta_k\} x\theta$ .

skin lesion images, then a sample with cases is selected at random from the pool of the images. To form a new tree, the selected skin lesion images are applied as the training sample. The optimal split on the images is done to divide the total node counts in the tree for images. For the forest growth, is maintained constant. As depicted in Eq. (39), the random forest ensemble method can be derived as the fusion of the different tree-structures. In (39), refers the classifier function, while be the linearly distributed random vector. In this work, each tree generates or outputs a unit vote

signifying the best class (the likelihood of belongingness of a sample to a specific class). Here, determines the tree formation value.

$$\{h(x, \theta_k), k = 1, 2, \dots, i \dots\} \quad (39)$$

The proposed method applied 80% samples (images) as training data, while balance 20% sample was considered to be the out-of-bag samples, which were applied to perform inner cross-validation to yield classification results. To be noted, in this research each of the base classifiers performs multi-class classification and annotates each input image as including Actinic Keratosis (AK), Basal Cell Carcinoma (BCC), Benign Keratosis-like Lesions (BKL), Dermatofibroma (DF), Melanoma (MEL), Nevus (NV) and Vascular Lesions (VASC) and accordingly labels them. The obtained score for each image is later applied to perform maximum voting-based ensemble.

#### 4. Extra tree classifier

It forms a cluster constituting unpruned decision trees as per the traditional top-down approach. Unlike aforesaid RF learner, ETC ensemble model contains randomization of attribute and cut-point selection while performing node-split task. This method makes ETC capable to constitute overall randomized trees embodying structures which are independent of the outputs of the training data. This method separates itself from the other tree-based ensemble like RF due to the two predominant factors. The first factor is that it divides nodes by selecting cut-points randomly, while the second factor is that it makes use of the overall training data for forest growth. The final predicted results from the deployed trees are used to generate and provide final predicted results by applying maximum voting ensemble (MVE) technique. Noticeably, the predominant method behind the proposed ETC technique is that the overall randomization of the cut-point and elements together by applying ensemble averaging method that reduces the variance in more effective manner in comparison to the weaker randomization techniques used in other machine learning methods. The use of images as training data minimizes the probability of bias and therefore yields higher accuracy towards the targeted skin cancer classification. Same as RF ensemble, it annotates each input skin lesion image as including Actinic Keratosis (AK), Basal Cell Carcinoma (BCC), Benign Keratosis-like Lesions (BKL), Dermatofibroma (DF), Melanoma (MEL), Nevus (NV) and Vascular Lesions (VASC), and assigns corresponding score, which is later applied by MVE to make final prediction decision.

#### 5. XGBoost ensemble

XGBoost is an extreme gradient boosting ensemble classifier, which is a distributed gradient boosted decision tree library, often applied for the large data learning and classification. In function, it yields parallel tree boosting concept, in which the prediction result of each decision tree algorithm serves as vote to make the final prediction. The capability to enhance gradient boosting, especially over the non-linear data space makes XGBoost robust for real-time classification with high accuracy and reliability. The aforesaid ability also makes it superior over the classical gradient boosting classifiers. It also possesses the ability to learn and classify large non-linear features without undergoing premature convergence and the global optimal centric swift convergence makes it suitable for varied run-time learning and classification tasks. Considering aforesaid strengths, we applied XGBoost algorithm as a base classifier for skin-cancer classification.

Let, the input feature possesses  $n$  instances signifying the  $m$  features  $X = \{x_i\}_{i=1}^n$ ,  $x_i \in R^m$  having labels  $y = \{y_i\}_{i=1}^n$ ,  $y_i \in \{\omega_j \in (1, 2, \dots, c)\}$ , where  $\omega_j$  be the  $j$ -th class from  $C$  total classes. In this case, an ensemble of the decision tree applying  $K$  additive functions perform learning and prediction, can be derived as per the equation (40)

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i), f_k \in F \quad (40)$$

In (40),  $F = \{f(x) = w_q(x)\} (q: R^m \rightarrow T, w \in R^T)$  signifies the space of the CART, while  $q$  and  $T$  represents the structure and the total number of leaves present within the tree. Each tree  $f_k$  belongs to a pattern  $q$  and leaf weight  $w$ . Now, for an input skin lesion image, it employs decision rules within the tree structure, given by  $q$  to classify it into leaves, which is then followed by the estimation of the final prediction result by adding the score in the corresponding leaves  $w$ . The regularization term (41) is used to learn the set of functions used for the ensemble learner.

$$\mathcal{L} = \sum_{i=1}^n l(\hat{y}_i, y_i) + \sum_{k=1}^K \Omega(f_k), \text{ where } \Omega(f_k) = \xi T + \frac{1}{2} \xi w^2 \quad (41)$$

In (41),  $l$  states a differential convex loss function which measures the disparity between the prediction  $\hat{y}_i$  and the target value  $y_i$ . The function  $\Omega(f)$  states the complexity of the tree  $f_k$ . In Eq. (41),  $\xi T$  and  $\xi w^2$  penalize each tree and extreme weights, correspondingly. The Eq. (41) contains the parametric functions which can't be enhanced by using classical optimization methods like the Euclidean space. Yet, due to the additive training characteristics, the objective function for the  $t$  iteration in the form of the



prediction at the previous iteration  $t - 1$  can be tuned by using the updated tree  $f_k$ :

$$\ell^{(t)} = \sum_{i=1}^n l\left(\hat{y}_i^{(t-1)} + f_t(x_i), y_i\right) + O(f_t) \quad (42)$$

Now, using Taylor expansion mechanism on (42), on the basis of the 1st and 2nd -order gradients (with reference to a cost function), a simplified objective function is derived (43).

$$\ell^{(t)} \approx \sum_{i=1}^n \left[ g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + O(f_t) \quad (43)$$

In (43),  $g_i = \delta_{\hat{y}_i^{(t-1)}} l\left(y_i, \hat{y}_i^{(t-1)}\right)$  and  $h_i = \delta_{\hat{y}_i^{(t-1)}}^2 l\left(y_i, \hat{y}_i^{(t-1)}\right)$

A DC algorithm predicts fixed values within a leaf, and therefore the tree  $f_k(x)$  can be defined as per  $w_q(x)$ , where  $w$  be the score vector for each leaf. Here,  $q(x)$  maps the data instance  $x$  to a leaf. Expanding the second term of (43), a sum over the tree leaves is obtained, where the regularization outputs are defined as per the equation (44).

$$\ell^{(t)} \approx \sum_{j=1}^T \left[ G_j w_j + \frac{1}{2} (H_j + \lambda) w_j^2 \right] + \lambda T, \text{ where } G_j = \sum_{i \in I_j} g_i, H_j = \sum_{i \in I_j} h_i \quad (44)$$

In (44),  $I_j = \{i | q(x_i) = j\}$  signifies the instance at leaf  $j$ . For a fixed structure tree, the cost function is reduced as  $\frac{d\ell^{(t)}}{dw_j} = G_j + (H_j + \lambda) w_j = 0$ , and the best weight of leaf node  $j$  is calculated as per the Eq. (45).

$$w^* = -\frac{G_j}{H_j + \lambda} \quad (45)$$

Now, substituting (45) into (44), the objective function to measure the best tree structure becomes (46).

$$\ell^{(t)} = -\frac{1}{2} \sum_{j=1}^T \frac{G_j^2}{H_j + \lambda} + \lambda T \quad (46)$$

In this work, the Eq. (46) was used to assess the split candidates in XGBoost learner, where to achieve the best split, the exact greedy algorithm and the global and local variant approximation method were applied over all the feasible splits pertaining to the input features. Though, the local approximation methods need smaller number of candidates than the global algorithm, the performance by the global method remains same as that of the local method. We used distributed gradient boosting driven XGBoost learner as base classifier, and thus the XGBoost method supports the exact greedy method for a single machine set and both local and global variant approximation techniques for all

settings. In XGBoost method, DART booster was applied that drops  $k$  trees during the  $m$ -th training round. Let,  $D = \sum_{k \in K} F_k$  be the leaf scores of the dropped trees and  $F_m = \eta F_m$  refers the leaf scores of a new tree; then, the cost function as derived in (42) can be redefined as per the Eq. (47).

$$\ell^{(t)} = \sum_{i=1}^n l\left(\hat{y}_i^{(m-1)} - D_i + F_m, y_i\right) + O(F_m) \quad (47)$$

In (47),  $D$  and  $F_m$  state the overshooting parameters, which are normalized, where XGBoost supports tree- and forest-based normalization method was taken into consideration. In this manner, the proposed XGBoost method classified each input skin lesion as Actinic Keratosis (AK), Basal Cell Carcinoma (BCC), Benign Keratosis-like Lesions (BKL), Dermatofibroma (DF), Melanoma (MEL), Nevus (NV) and Vascular Lesions (VASC) and assigns corresponding score, which is later applied by the MVE ensemble to perform final (skin-cancer) prediction.

Once obtaining the predicted score by each of the base classifier, the proposed MVE ensemble method estimates the highest score and annotates the input image for the best possible skin-cancer class (i.e., Actinic Keratosis (AK), Basal Cell Carcinoma (BCC), Benign Keratosis-like Lesions (BKL), Dermatofibroma (DF), Melanoma (MEL), Nevus (NV) and Vascular Lesions (VASC)). The simulation results and allied inferences are given in the subsequent section.

## Results and discussion

The last few years have witnessed high-pace rise in skin cancer cases globally. On the other hand, the diversities and ambiguities over the different kinds of skin cancer cases such as the Actinic Keratosis (AK), Basal Cell Carcinoma (BCC), Benign Keratosis-like Lesions (BKL), Dermatofibroma (DF), Melanoma (MEL), Nevus (NV) and Vascular Lesions (VASC) make it difficult to diagnose specific cancer type and make suitable clinical decisions. Human assessment based on visual analysis is often prone to the (human) error and hence vision-computing has emerged as a viable approach. Amongst the major state-of-arts designed towards skin-cancer detection and classification, deep learning methods have performed superiorly. The ability to extract and learn fine-grained features from the input dermatological images makes deep-learning solutions more effective; however, the at hand data challenges such as the data-imbalance, skewed learning, gradient vanishing, gradient explode, accuracy degradation, and lack of contextual details confine majority of such state-of-arts. Those

methods designed based on the CNN and traditional DNN variants are often exploiting local hierarchical features and lack contextual details that might impact overall learning efficiency and reliability of the solution. On the other hand, these existing methods processes feature extraction over complete dermatological images encompassing both ROI (i.e., skin-lesions) and non-ROI. In real-world scenario such data can be skewed with non-ROI as majority class and therefore extracting non-ROI details for further training can not only impose computational costs but can also skew the learning and prediction results. Consequently, it can impact overall performance and its reliability. Such challenges can be severe over the multi-class classification problem, where one intends to design a fit-to-all solution to detect and classify skin cancer lesions. In addition to this the majority of the state-of-arts fail to consider ROI-specific details for feature extraction and learning. To alleviate this, the ROI segmentation (i.e., skin lesion) can be considered, which can then be followed by ROI-specific feature extraction and learning. It can not only improve the feature informativeness but can also address instance-level class-imbalance that eventually can improve overall learning and prediction accuracy. In addition to this the use of the different feature models encompassing the residual deep networks, transfer learning models and spatio-textural feature models can yield information rich and heterogenous feature environment to serve optimal learning and prediction. Unlike traditional researches where the authors have applied stand-alone machine learning classifiers to perform prediction, the use of maximum voting ensemble (MVE) learning concept, specially designed with the base classifiers pertaining to the different pattern mining can yield higher accuracy. It can be effective towards at hand multi-class skin cancer detection and classification purposes.

Considering above stated research challenges and allied scopes, in this paper semantic segmentation assisted deep ensemble feature learning model is proposed and developed for (multi-class) skin-cancer detection and classification (SDENet). In this work, HAM10000 datasets embodying ISIC2019 dermatological images were taken into consideration. Functionally, the SDENet at first executed different pre-processing methods such as the intensity equalization, histogram equalization, z-score normalization and resizing in sequence. Additionally, to address sample level class-imbalance problem, synthetic minority over-sampling (SMOTE) was performed with the data augmentation technique to enable fair sample distributions across the skin-cancer images (in each type). The pre-processed dermatological images were then processed for the semantic segmentation task so as to retain skin-lesion detection and segmentation (say, ROI). To achieve it, the proposed SDENet model applied firefly heuristic driven Fuzzy C-means clustering

(FFCM) method that not only enabled automated skin-lesion segmentation but also avoided manual seed point definition or threshold estimation challenges, as witnessed in the traditional methods like K-Means clustering, FCM and Otsu method. To implement FFCM, the Firefly algorithm was designed with the initial generation of 25, while the initial population size was fixed at 30. The brightness coefficient was fixed at 1. Once segmenting the ROI-specific skin-lesions, the channel-wise multiplication was done with the original images to yield ROI-specific RGB color images, which was then passed as input to the proposed deep spatio-textural ensemble feature extraction and fusion (DeS-TEFF) model. To ensure sufficiently large feature heterogeneity, the proposed SDENet model used the different feature models including AlexNet, DenseNet121 and GLCM models. Here, AlexNet and DenseNet121 served deep features, while GLCM provided spatio-textural feature distribution for each ROI-specific dermatological image. Noticeably, the AlexNet deep model provided 4096-dimensional information rich features, while DenseNet121 provided residual feature for further learning and prediction. Unlike classical CNN or RNN methods, DenseNet121 due to its layer-wise learning ability and feature reuse served information rich feature even with the reduced hyperparameters, which can improve computational efficiency. Both the AlexNet and DenseNet121 deep models were designed with the ADAM non-linear adaptive learning model, where the number of epochs was fixed at 100. The initial learning rate was fixed at 0.0001, while the batch size was fixed at 8. Once extracting AlexNet, DenseNet121 and GLCM features (energy, entropy, correlation, contrast, homogeneity, mean, standard deviation, variance, kurtosis and skewness), horizontal concatenation was performed to yield composite feature set, which was later processed for PCA feature selection and z-score normalization. These feature engineering methods helped to alleviate the local minima and convergence, and over-fitting problems. The normalized features were then trained by means of the proposed Heterogenous Ensemble Classifier, embodying SVM, DT, k-NN, Naïve Bayes, Random Forest, Extra Tree Classifier and XGBoost classifiers. Being a MVE ensemble method, each of the base classifier yielded individual prediction results and corresponding score for each input image, which is then processed by maximum score estimation. Thus, with the highest core observed the proposed MVE ensemble method predicted final class for each input skin-lesion image and thus classified each dermatological images in the different categories (i.e., Actinic Keratosis (AK), Basal Cell Carcinoma (BCC), Benign Keratosis-like Lesions (BKL), Dermatofibroma (DF), Melanoma (MEL), Nevus (NV) and Vascular Lesions (VASC)).

**Table 3** Performance parameters

| Parameter | Mathematical Expression                                   |
|-----------|-----------------------------------------------------------|
| Accuracy  | $\frac{(TN+TP)}{(TN+FN+FP+TP)}$                           |
| Precision | $\frac{TP}{(TP+FP)}$                                      |
| Recall    | $\frac{TP}{(TP+FN)}$                                      |
| F-Measure | $2 \cdot \frac{Recall \cdot Precision}{Recall+Precision}$ |

The overall proposed model was developed by using MATLAB 2022b software simulation tool, where the different toolboxes including signal processing, image processing, deep learning and ML were taken into consideration. The proposed model was simulated over a central processing unit (CPU) armored with Microsoft Office operating system, 16 GB RAM (memory), functional at 3.2 GHz Intel-i5 processor.

To assess efficacy of the proposed VF prediction model, the classification framework resulted confusion matrix in terms of the true positive (TP), true negative (TN), false positive (FP) and false negative (FN). The different statistical performance parameters were obtained, which are defined in Table 3.

In reference to the research questions (RQ) framed in the Section III, this work hypothesizes that:

**H01** *The data augmentation followed by the FFCM segmentation can achieve more accurate skin-cancer detection and classification solution.*

**H02** *The use of deep spatio-textural ensemble feature extraction and fusion (DeS-TEFF) model encompassing GLCM, AlexNet and DenseNet121 feature models can serve sufficiently large feature set as input to enable highly accurate skin-cancer classification solution.*

**H03** *The use of heterogenous ensemble learning classifier encompassing SVM, DT, RF, ETC and XGB algorithms with the proposed DeS-TEFF feature model yield higher accuracy towards skin lesion detection and classification.*

**H04** *The strategic amalgamation of the FFCM segmentation (RQ1), ROI-specific DeS-TEFF feature extraction (i.e., GLCM, AlexNet and DenseNet121) (RQ2), PCA feature selection, z-score normalization and HEL ensemble (RQ3) be effective to serve a robust, reliable and scalable skin-cancer detection and classification solution.*

In reference to these RQs and hypotheses, the overall performance characterization is done in terms of the intra-model assessment and inter-model assessment. Here, in intra-model assessment the efficiency is measured over the

**Table 4** Segmentation model analysis

| Features/Model       | Accuracy (%) | Precision (%) | Recall (%) | F-Measure |
|----------------------|--------------|---------------|------------|-----------|
| Without Segmentation | 94.68        | 95.11         | 97.92      | 0.96      |
| With FFCM            | 98.97        | 99.38         | 98.94      | 0.99      |

different feature models, different machine learning classifiers and with the FFCM segmentation distinctly to assess whether the proposed method performs as expected. In other words, the intra-model assessment enables to identify the best performing feature model, classifier etc. so as to generalize the best-performing solution. On the contrary, in the inter-model assessment the performance is compared with the other state-of-art skin cancer detection and classification solutions. The discussion of the overall simulation results and allied inferences is given in the subsequent sections.

## 1. Intra-Model assessment

In this performance characterization, the simulations were made with FFCM and without FFCM segmentation, where in the last case, the whole input images from the HAM10000 ISIC2019 datasets were passed as input to the GLCM, AlexNet and DenseNet121 feature extraction models. On the contrary, with FFCM segmentation only the ROI-specific (RGB) color images were passed to the aforesaid feature models. Here, HEL ensemble learning model was applied as a classifier. On the other hand, the simulations were made with the different feature models, such as the GLCM, AlexNet and DenseNet121, and combination of these feature models (i.e., GLCM+AlexNet+DenseNet121). Finally, the classification was done by applying base classifiers as standalone algorithm as well as the proposed HEL-based MVE ensemble. The simulation results obtained and allied inferences are discussed in the subsequent sections.

As depicted in Table 4, the average accuracy without segmentation-based solution is obtained as 94.68%, while the proposed FFCM-based segmentation and subsequent ROI-specific feature learning method yields 98.97%, which is higher, signifying its superiority towards run-time environment. To be noted, for without segmentation realization the complete original image was passed as input to the proposed feature model (i.e., DeS-TEFF). On the contrary, with FFCM segmentation-based solution the ROI-specific color images were passed to the DeS-TEFF for feature extraction and learning. In both data cases, MVE-HLE ensemble classifier was applied. Observing the results (Figs. 6 and 7 and Table 4), it can be found that the classical approaches (i.e., without segmentation) yields skin-cancer classification precision, recall and F-Measure of 95.11%, 97.92% and 0.96, respectively. On the other hand, the proposed

FFCM segmentation model over HAM10000 (ISIC2019) dataset yielded multi-class skin-cancer classification accuracy, precision, recall and F-Measure of 98.97%, 99.38%, 98.94% and 0.99, respectively (Fig. 6). It clearly confirms that the ROI specific feature learning can yield superior performance than without ROI segmentation-based solutions.

To assess whether the proposed MVE-HEL ensemble learning method yields superior skin-cancer prediction accuracy or not, the simulations were made with the different base-classifiers as well as the proposed MVE-HEL classifier. Noticeably, in this assessment, the proposed DeS-TEFF feature model (i.e., FFCM-segmentation driven ROI-specific deep spatio-textural feature ensemble) was considered as input feature. Additionally, these machine learning classifiers were trained over the different features separately (i.e., GLCM, AlexNet, DenseNet121 and composite DeS-TEFF features). The simulation results obtained are given in Table 5 and 6, and Figs. 8 and 9.

The simulation results confirmed that the GLCM features yield skin-cancer prediction accuracy of 95.22%, while AlexNet yielded 95.16% of multi-class skin-cancer prediction. The DenseNet121 as standalone feature model yielded the average accuracy of 95.25%. On the contrary, the proposed DeS-TEFF model (say, proposed SDENet) yielded the average prediction accuracy of 96.45%, which is higher than the other feature models. The F-Measure performance too confirmed that the GLCM method yielded the average F-Measure of 0.94. Similarly, AlexNet, DenseNet121 and SDENet models yielded the F-measures of 0.94, 0.95 and 0.97, respectively. Thus, the overall results confirm that the proposed SDENet and corresponding deep spatio-textural feature model (i.e., DeS-TEFF) outperforms other feature models.

To assess the relative efficiency of the different machine learning models, the simulations were made with the different machine learning classifiers and the proposed MVE-HEL ensemble models. The simulation results reveal that the SVM classifier shows the average accuracy of 94.08% towards multi-class (skin-cancer) classification. The DT algorithm, RF, ETC and XGB algorithm on the other hand show the skin-cancer detection and classification accuracy of 94.16%, 96.32%, 95.02% and 96.35%. The proposed MVE-HEL method shows the average skin-cancer prediction accuracy of 96.71%. The simulation results also reveal that the SVM, DT, RF, ETC, XGB and MVE-HEL models shows the F-Measure values of 0.94, 0.94, 0.95, 0.94, 0.96 and 0.97, respectively. The overall results confirm that that proposed MVE-HEL model outperforms other machine learning algorithms and hence serves a robust solution towards skin-cancer detection and classification tasks are given in Table 7 and Figs. 10 and 11.

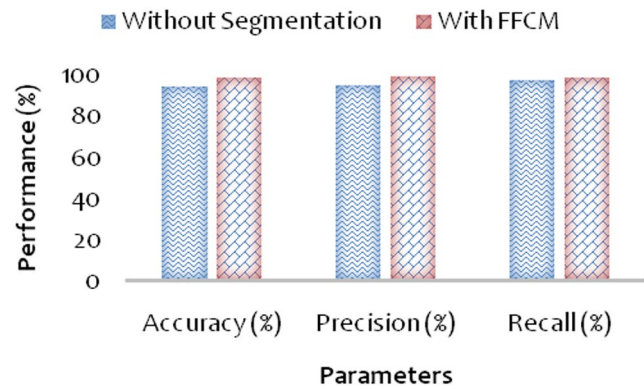


Fig. 6 Accuracy, precision and recall with FFCM segmentation

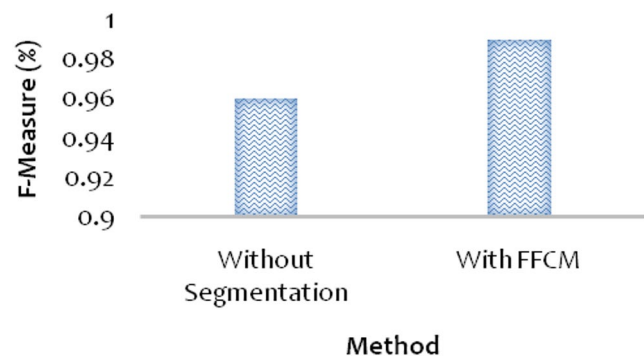


Fig. 7 F-Measure performance with FFCM segmentation

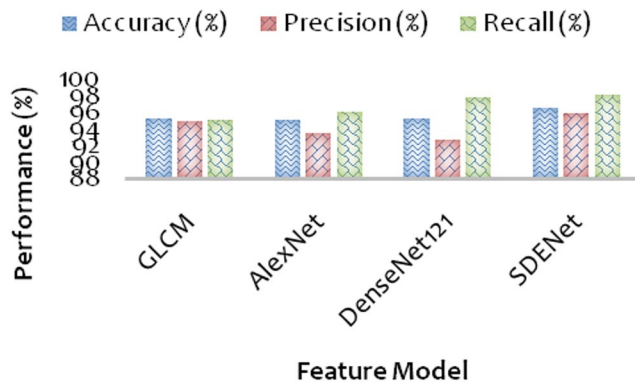
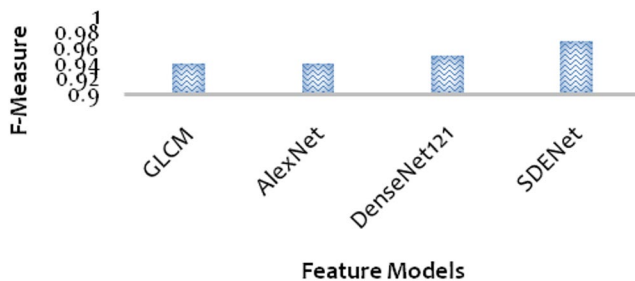
Table 5 Performance of the different feature models

| Features/Model    | Machine Learning | Accuracy (%) | Precision (%) | Recall (%) | F-Measure |
|-------------------|------------------|--------------|---------------|------------|-----------|
| GLCM              | SVM              | 94.89        | 95.21         | 95.00      | 0.95      |
|                   | DT               | 94.00        | 93.96         | 93.92      | 0.94      |
|                   | RF               | 95.34        | 95.33         | 96.21      | 0.96      |
|                   | ETC              | 95.30        | 93.96         | 93.92      | 0.93      |
|                   | XGB              | 96.80        | 95.95         | 96.54      | 0.96      |
|                   | HEL-MVE          | 95.03        | 94.97         | 94.82      | 0.95      |
| AlexNet           | SVM              | 93.97        | 91.04         | 98.00      | 0.93      |
|                   | DT               | 93.01        | 91.21         | 99.02      | 0.93      |
|                   | RF               | 96.66        | 96.02         | 96.85      | 0.96      |
|                   | ETC              | 95.33        | 94.10         | 88.03      | 0.94      |
|                   | XGB              | 96.38        | 95.19         | 95.93      | 0.95      |
|                   | HEL-MVE          | 95.63        | 92.99         | 98.11      | 0.95      |
| DenseNet121       | SVM              | 93.27        | 91.19         | 98.21      | 0.93      |
|                   | DT               | 93.93        | 87.00         | 99.03      | 0.93      |
|                   | RF               | 97.72        | 94.37         | 96.57      | 0.95      |
|                   | ETC              | 94.12        | 93.06         | 98.22      | 0.93      |
|                   | XGB              | 95.24        | 91.11         | 99.00      | 0.96      |
|                   | HEL-MVE          | 97.24        | 99.69         | 95.82      | 0.98      |
| Proposed DeS-TEFF | SVM              | 94.21        | 91.11         | 99.20      | 0.95      |
|                   | DT               | 95.70        | 94.40         | 98.89      | 0.97      |
|                   | RF               | 97.48        | 96.01         | 96.13      | 0.96      |
|                   | ETC              | 95.34        | 95.33         | 96.21      | 0.96      |
|                   | XGB              | 97.01        | 98.82         | 99.38      | 0.99      |
|                   | HEL-MVE          | 98.97        | 99.38         | 98.94      | 0.99      |



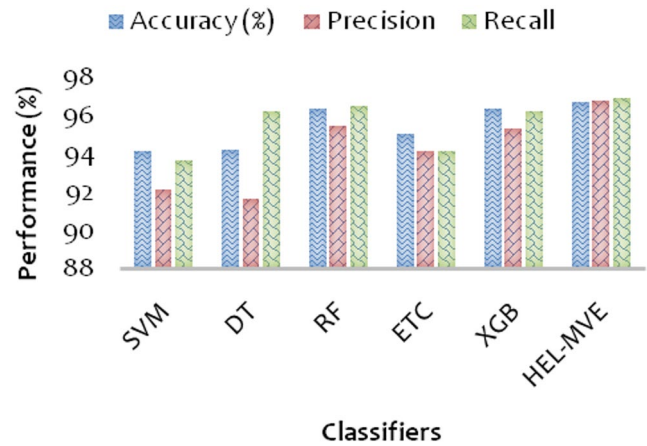
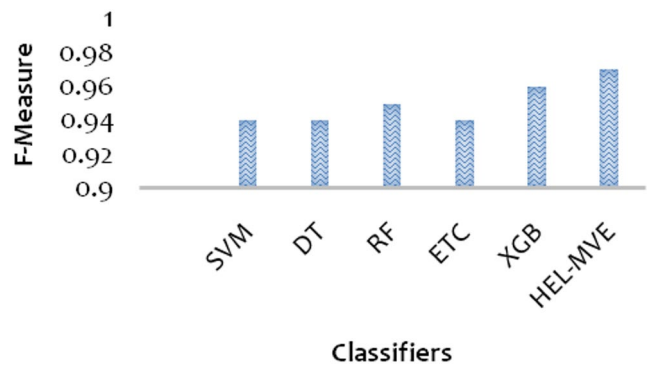
**Table 6** Average performance of the different feature models

| Feature Model | Accuracy (%) | Precision (%) | Recall (%) | F-Measure |
|---------------|--------------|---------------|------------|-----------|
| GLCM          | 95.22        | 94.89         | 95.06      | 0.94      |
| AlexNet       | 95.16        | 93.42         | 95.99      | 0.94      |
| DenseNet121   | 95.25        | 92.73         | 97.8       | 0.95      |
| SDENet        | 96.45        | 95.84         | 98.12      | 0.97      |

**Fig. 8** Accuracy, precision and recall with the different feature models**Fig. 9** F-Measure performance with the different feature models**Table 7** Performance of the different machine learning classifiers

| Machine Learning | Accuracy (%) | Precision | Recall | F-Measure |
|------------------|--------------|-----------|--------|-----------|
| SVM              | 94.08        | 92.13     | 93.6   | 0.94      |
| DT               | 94.16        | 91.64     | 96.21  | 0.94      |
| RF               | 96.32        | 95.43     | 96.44  | 0.95      |
| ETC              | 95.02        | 94.11     | 94.09  | 0.94      |
| XGB              | 96.35        | 95.26     | 96.21  | 0.96      |
| MVE-HEL          | 96.71        | 96.75     | 96.92  | 0.97      |

Observing overall results and allied inferences, it can be confirmed that the FFCM segmentation can achieve more accurate skin-cancer detection and classification solution. It affirms acceptability of the hypothesis H01. It also confirms acceptance of the proposed RQ1 research question. The results also confirm that the proposed DeS-TEFF model encompassing GLCM, AlexNet and DenseNet121 feature models can serve sufficiently large feature set as input to enable highly accurate skin-cancer classification solution. It confirms acceptance of the research hypothesis H02. Similarly, the results confirm that the heterogenous ensemble learning classifier encompassing SVM, DT, RF, ETC and

**Fig. 10** Accuracy, precision and recall performance with the different machine learning classifiers**Fig. 11** F-Measure performance with the different machine learning classifiers

XGB algorithms with the proposed DeS-TEFF feature model yield higher accuracy towards skin lesion detection and classification. It confirms the acceptance of the RQ3 and H03.

## 2. Inter-model assessment

To assess efficiency of the proposed SDENet model, its performance was compared with the other state-of-arts. To examine the relative performance accuracy was considered as a common parameter. The different techniques and corresponding performance outputs are given in the Table 8.

Comparative evaluations with over 29 state-of-the-art models demonstrate that SDENet consistently achieves higher classification accuracy and robustness, outperforming both conventional CNNs and recent hybrid frameworks on the HAM10000 dataset. The above results depict efficacy of the state-of-the art approaches designed towards skin cancer classification. Interestingly, despite the significant efforts a number of existing methods suffer from poor accuracy that confines their suitability towards run-time

applications. Even the deep learning methods such as the CNN [26, 40, 41, 72, 73, 75, 82], DNN and variants [19, 79, 81], residual networks [20, 31, 38, 81] have shown poor performance. The depth assessment reveal that the CNN deep models as the feature extraction technique has shown the skin-cancer classification accuracy of 88.02% [82], 92.20% [72], 94.87% [73], 85.50% [68], 94.52% [75] etc. Undeniably, CNN features which are nothing else but the local hierarchical features have undergone reduced performance, especially with the traditional machine learning classifiers such as the SVM algorithm [68]. Though, the same feature (i.e., CNN) with RF ensemble classifier has shown improved skin-cancer classification accuracy [75]. Despite the inference that the CNN lacks contextual details and hence undergoes reduced performance, the use of RF ensemble has helped to achieve improved performance in the state-of-art methods. Residual network such as the ResNet and DenseNet have also shown superior efficiency (98.73%) than the CNN or DNN variants (MobileNet 98.39%). However, DenseNet could have achieved superior efficiency over the ResNet with the improved structure. Since in this work the proposed SDENet model intended to exploit DenseNet121, which is considered to yield not only tuned features but also with the reduced computation has enabled achieving skin cancer classification accuracy of 98.97%, which is higher than the traditional methods. Moreover, the depth quantification of the state-of-arts reveal that the skin-lesion segmentation assisted ROI-specific feature learning performs superior [58, 60]. Yet, the classical standard threshold-based methods like OTSU methods [53, 58] have shown poor accuracy (80% [53]). Though, improving the same with the Fuzzy based optimization enabled achieving higher accuracy (i.e., 94% [60]). Yet, merely exploiting color, shape and texture features over the large complex histopathological images can't guarantee reliability of the solution over the run-time environment where the patients or subjects can have the difference in ethnicity or demographics. It clearly demands amalgamation of more feature sets such as the GLCM (92.10% [50]). Unlike traditional efforts [50], where merely GLCM feature was applied to perform skin-cancer classification, in this work a composite feature model was applied that not only embodied GLCM as one of the features but also amalgamated DenseNet121 and AlexNet features together to achieve highly reliable learning and prediction. To be noted, there exists no significant effort which could have considered end-to-end challenges such as the data imbalance, ROI specific feature learning, deep spatio-textural feature learning and voting based ensemble, which could have achieved highly accurate performance towards run-time decisions. On the contrary, in this work a robust approach was designed that amalgamated image augmentation or resampling, ROI-specific feature

extraction and learning, deep spatio-textural feature learning and MVE ensemble. This ability enabled the proposed SDENet model yielding the skin-cancer classification accuracy of 98.97%, precision of 99.38%, recall 98.94% and F-Measure of 0.99. The overall performance confirms that the proposed model outperforms other approaches with more reliability and efficacy. Noticeably, the methods such as the [27], in which at first UNet was applied to segment skin lesions followed by the DenseNet77 feature extraction and learning yielded the accuracy of 99.21%. Similarly, the approach proposed in [62] which amalgamated MobileNet and SqueezeNet yielded the highest accuracy of 99.13%, which is almost 0.2% higher than the SDENet; however, it can fall pray to the computational challenges such as the local minima and convergence. Moreover, the performance diversity over the non-linear data or feature space can't be ruled out. On the contrary, the proposed model due to the MVE-HEL learning enabled achieved more reliable (due to the consensus driven prediction solution) towards run-time environment. It confirms affirmative acceptance of RQ4 and hence H04. The originality of the proposed SDENet framework lies in its integration of semantic segmentation-assisted feature learning with a heterogeneous ensemble architecture. Unlike conventional ensemble or hybrid models that combine CNN outputs without region-specific optimization, SDENet employs Firefly-optimized Fuzzy C-Means (FFCM) to extract precise lesion regions before deep feature learning. This design ensures that feature fusion is both region-aware and diagnostically relevant. Furthermore, the DeS-TEFF module fuses AlexNet and DenseNet121 features, which represent complementary hierarchical representations—AlexNet capturing broader texture distributions and DenseNet121 learning fine-grained semantic cues—along with GLCM-based texture descriptors. The inclusion of PCA-driven redundancy minimization and a heterogeneous voting ensemble provides a multi-perspective decision space that extends beyond the performance of standard CNN ensembles. This integrative approach advances the reliability and interpretability of skin cancer CAD systems, aligning with the methodological rigor demonstrated in comparative diagnostic frameworks such as the study on breast cancer detection [94]. The strong classification performance of SDENet indicates its potential clinical value, yet its broader utility depends on robustness across diverse imaging and demographic conditions. Variations in illumination, device resolution, and skin tone can affect lesion appearance; however, the FFCM-based segmentation and DeS-TEFF fusion enhance generalization by focusing on discriminative, region-specific features. Future studies will validate SDENet on multi-institutional and multi-ethnic datasets to assess adaptability. This aligns with

**Table 8** Inter-model assessment

| Method | Technique                                         | Accuracy (%)        |
|--------|---------------------------------------------------|---------------------|
| [1]    | Optimized CNN                                     | 96.10               |
| [5]    | DeepMetaForge                                     | 97.50               |
| [12]   | Multimodal Analysis                               | 96.30               |
| [19]   | MobileNet and DenseNet201                         | 95.50               |
|        | MobileNetv2                                       | 91.86               |
| [20]   | ResNet50                                          | 83.70               |
|        | InceptionV3                                       | 85.80               |
| [21]   | AlexNet+ SVM                                      | 95.10               |
| [22]   | CNN                                               | 83.00               |
| [23]   | CNN                                               | 65.50               |
| [24]   | ResNet-18                                         | 94.47               |
| [25]   | AlexNet                                           | 91.00               |
| [26]   | CNN                                               | 88.00               |
| [27]   | DenseNet77-driven UNet                            | 99.21               |
| [28]   | Xception,<br>ResNet50,<br>ResNet101, and<br>VGG16 | 97.41               |
| [29]   | SqueezeNet                                        | 89.20               |
| [35]   | InSiNet                                           | 96.10               |
| [38]   | UNet+ DenseNet                                    | 84.00               |
| [40]   | CNN                                               | 78.00               |
| [41]   | CNN- VGG16                                        | 88.00               |
| [43]   | GLCM+ HOG+ SVM                                    | 97.80               |
| [47]   | MobileNet, and<br>VGG-16                          | 81.52<br>80.07      |
| [48]   | Retina Net+ CRF                                   | 94.20               |
| [49]   | DNN                                               | 93.06               |
| [52]   | Canny Edge Detector+ ABCD Features                | 93.60               |
| [53]   | Otsu+ SIFT                                        | 80.00               |
| [54]   | ResNet50                                          | 94.00               |
| [55]   | HOG+ SURF                                         | 94.92               |
| [56]   | CNN+ SVM                                          | 82.60               |
| [57]   | GLCM+ PCA+ SVM                                    | 92.10               |
| [58]   | OTSU+ HOG+ SVM                                    | 96.30               |
| [59]   | OTSU+ Color, shape and texture+ DT                | 85.00               |
| [60]   | Fuzzy-OTSU+ Shape Color and texture+ ANN          | 94.00               |
| [62]   | MobileNet+ SqueezeNet                             | 99.13               |
| [63]   | FCM+ MMLVR-WT                                     | 95.76 (sensitivity) |
| [68]   | CNN+ SVM                                          | 85.50               |
| [70]   | AlexNet, ResNet-18, and VGG-16                    | 97.55               |
| [71]   | VGG-16                                            | 78.00               |
| [72]   | CNN                                               | 92.20               |
| [73]   | CNN                                               | 94.87               |
| [74]   | VGG16, ResNet50, and Inception-v3                 | 88.10               |
| [75]   | CNN+ RF                                           | 94.52               |
| [76]   | MobileNet+ RF                                     | 86.50               |
| [79]   | DNN+ ELM                                          | 93.40               |
| [31]   | DenseNet                                          | 92.50               |
|        |                                                   | (AUC)               |
| [81]   | ResNet                                            | 98.73               |
|        | VGG16                                             | 99.76               |
|        | MobileNet16                                       | 98.39               |
| [82]   | CNN                                               | 88.02               |
| [83]   | SCDNet (VGG16+ CNN)                               | 90.48               |

recent discussions on model robustness and translational potential in heterogeneous clinical data environments [95].

## Conclusion

In this paper, a robust semantic segmentation assisted deep ensemble feature learning model was proposed for skin-cancer detection and classification (SDENet). Considering data specific run-time challenges and allied computational complexities, the SDENet model intended to optimize multiple aspects including data resampling, semantic and automatic segmentation, ROI-specific feature extraction and fusion, and ensemble consensus-driven classification. Here, key motive was to ensure reliability and generalizability of the proposed CAD solution towards real-time skin-cancer detection and classification. Noticeably, unlike traditional binary-class classification problems, where the state-of-arts merely classify input dermoscopic images as benign and malignant or benign and melanoma, this research focused on performing multi-class classification to classify input dermoscopic images in the different categories including Actinic Keratosis (AK), Basal Cell Carcinoma (BCC), Benign Keratosis-like Lesions (BKL), Dermatofibroma (DF), Melanoma (MEL), Nevus (NV) and Vascular Lesions (VASC). Undeniably, the skin lesions spatio-textural patterns and features for these different types of skin-cancer types are different, complex and ambiguous, and therefore, the proposed SDENet model focused to improve skin lesion segmentation, segmented ROI specific deep spatio-textural feature extraction and fusion, feature selection, normalization and ensemble-classification. Functionally, the SDENet model at first performed pre-processing tasks over the input HAM10000 dermoscopic dataset, where intensity equalization, histogram equalization, resizing and z-score normalization was performed. Subsequently, it executed synthetic minority over-sampling (SMOTE) technique over each class image samples to alleviate any probable class-imbalance or skewed data condition. Undeniably, the use of SMOTE could address the sample-level class-imbalance; however, in real-world dermatological images the instance or pixel level imbalance can't be ruled out. In other words, in real-world data conditions the skin-lesion sizes can be relatively smaller than the non-ROI or background regions. Extracting and learning features over these imbalanced images can impact overall learning and prediction results. To alleviate it, the SDENet model applied Firefly heuristic driven FCM clustering model which automatically segmented skin-lesion area without demanding manual seed-points or centroid information, as seen in normal FCM-clustering or K-means clustering-based solutions. It not only addressed data imbalance conditions, but also

served automatic ROI-specific feature extraction and learning to achieve higher accuracy. The ROI-specific images are then processed for deep spatio-textural ensemble feature extraction and fusion (DeS-TEFF). The proposed SDENet model applied AlexNet, DenseNet121 and GLCM features altogether for learning and prediction, which not only served sufficiently large and information rich features, but also alleviated the challenges like gradient vanishing, gradient explode, accuracy degradation etc., which is often observed in traditional CNN or DNN networks. Additionally, the amalgamation of GLCM features helped to introduce additional spatio-textural features that with the aforesaid deep ensemble features improved feature learning and prediction results. On the other hand, the PCA feature selection followed by z-score normalization helped to avoid local minima, convergence and over-fitting problems. Unlike traditional methods where the authors have applied standalone machine learning classifier to classify features, the proposed SDENet designed a heterogenous ensemble classifier embodying a total of seven classifiers, including SVM, DT, k-NN, Naïve Bayes, Random Forest, Extra Tree Classifier and XGBoost algorithms to perform maximum voting-based classification. The performance quantization revealed that the proposed SDENet model exhibits the average skin-cancer classification accuracy of 98.97%, precision 99.38%, recall 98.94% and F-Measure 0.99, confirming its superiority over other existing approaches for real-time skin cancer diagnosis purposes. The depth quantification also revealed that the proposed SDENet with the FFCM segmented ROI-specific feature yields almost (%) higher accuracy than the model without FFCM segmentation. On the other hand, the GLCM features, AlexNet, DenseNet121 yielded 95.03%, 95.63%, and 97.24% accuracies as the standalone feature model. On the contrary, their fusion resulted an improved accuracy of (%), exhibiting its efficacy and robustness towards the targeted skin-cancer detection and classification tasks. Finally, the accuracy obtained by the proposed ensemble learning model (i.e., MVE-HEL) was found superior 98.97% than the standalone machine learning classifiers (SVM (94.08%),DT (94.16%),RF (96.32%), ETC (95.02%) and XGB (96.35%)). The overall results confirm superiority of the proposed SDENet model over other state-of-art techniques for skin-cancer detection and classification. Being a multi-class classification solution, it can serve broader horizon for skin-cancer classification than the binary classification-oriented solutions. The overall results confirm real-time significance of the proposed model towards skin-cancer diagnosis for earlier clinical decisions. Undeniably, the proposed FFCM segmentation model with the GLCM, AlexNet and DenseNet121 features yielded superior efficiency; yet, the use of deep semantic techniques like UNet can also be of great significance due to its pixel-wise



classification and automatic segmentation capabilities over ambiguous edges. Moreover, other deep networks like the ResNet101, SqueezeNet etc. can be assessed towards feature extraction and learning. These key inferences can be considered as the future scopes.

**Author contributions** Ch. Srilakshmi contributed to the conception, design, implementation, experiments, and drafting of the manuscript. N. Ramakrishnaiah and E. Laxmi Lydia supervised the research, provided technical guidance, and refined the methodology.

## Declarations

**Conflict of interest** The authors declare no Conflict of interest.

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## References

- Mridha K, Uddin MM, Shin J, Khadka S, Mridha MF (2023) An interpretable skin cancer classification using optimized convolutional neural network for a smart healthcare System, in IEEE access 11:41003–41018
- Key Statistics for Melanoma Skin Cancer (2022) American Cancer Society, Atlanta, GA, USA
- Ankad B, Sakhare P, Prabhu M (2017) Dermoscopy of non-melanocytic and Pink tumors in brown skin: A descriptive study. *Indian J Der Matopathol Diagn Dermatol* 4(2):41
- Pai K, Giridharan A (2019) Convolutional neural networks for classifying skin lesions, in Proc. TENCON IEEE Region 10 Conf. (TENCON) 1794–1796
- Vachmanus S, Noraset T, Piyanonpong W, Rattananukrom T, Tuarob S (2023) DeepMetaForge: A deep Vision-Transformer Metadata-Fusion network for automatic skin lesion Classification, in IEEE access 11:145467–145484
- Su Q, Hamed HNA, Isa MA, Hao X, Dai X (2024) A GAN-Based Data Augmentation Method for Imbalanced Multi-Class Skin Lesion Classification, in IEEE Access 12:16498–16513
- Magdy A, Hussein H, Abdel-Kader RF, Salam KAE (2023) Performance enhancement of skin cancer classification using computer Vision, in IEEE access 11:72120–72133
- Goldstein BG, Goldstein AO (2001) Diagnosis and management of malignant melanoma. *Amer Family Physician* 63(7):1359–1369
- Olmez Y, Koca GO, Sengür A, Acharya UR, Mir H (2024) Improved PSO with visit table and multiple direction search strategies for skin cancer image Segmentation, in IEEE access 12:840–867
- Zhou X et al (May 2023) CUSS-Net: A cascaded Unsupervised-Based strategy and supervised network for biomedical image diagnosis and segmentation. *IEEE J Biomedical Health Inf* 27(5):2444–2455
- Sinz C et al (2017) Accuracy of dermatoscopy for the diagnosis of non-pigmented cancers of the skin, *J. Amer. Acad. Dermatol* 77(6):1100–1109
- Lyakhov PA, Lyakhova UA, Kalita DI (2023) Multimodal analysis of unbalanced dermatological data for skin cancer Recognition, in IEEE access 11:131487–131507
- Noronha SS, Mehta MA, Garg D, Kotecha K, Abraham A (2023) Deep Learning-Based Dermatological Condition Detection: A Systematic Review with Recent Methods, Datasets, Challenges, and Future Directions, in IEEE Access 11:140348–140381
- Saeed M, Naseer A, Masood H, Rehman SU, Gruhn V (2023) The power of generative AI to augment for enhanced skin cancer classification: A deep learning Approach, in IEEE access 11:130330–130344
- Qian X et al (2024) SPCB-Net: A Multi-Scale skin cancer image identification network using Self-Interactive attention pyramid and Cross-Layer Bilinear-Trilinear Pooling, in IEEE access 12:2272–2287
- Sathvika VSSBT et al (2024) Pipelined structure in the classification of skin lesions based on Alexnet CNN and SVM model with Bi-Sectional texture Features, in IEEE access 12:57366–57380
- Hoang V-D, Vo X-T, Jo K-H (2023) Categorical weighting domination for imbalanced classification with skin cancer in intelligent healthcare Systems, in IEEE access 11:105170–105181
- Dai W, Liu R, Wu T, Wang M, Yin J, Liu J (2024) Deeply supervised skin lesions diagnosis with stage and branch attention. *IEEE J Biomedical Health Inf* 28(2):719–729
- Ur Rehman MZ, Ahmed F, Alsuhbany SA, Jamal SS, Ali MZ, Ahmad J (2022) Classification of skin cancer lesions using explainable deep learning, *Sensors* 22(18):6915
- Gouda W, Sama NU, Al-Waakid G, Humayun M, Jhanjhi NZ (2022) Detection of skin cancer based on skin lesion images using deep learning, *Healthcare* 10(7):1183
- Dorj U-O, Lee K-K, Choi J-Y, Lee M (2018) The skin cancer classification using deep convolutional neural network. *Multimedia Tools Appl* 77(8):9909–9924
- Fu'adah YN, Pratiwi NC, Pramudito MA, Ibrahim N (2020) Convolutional neural network (CNN) for automatic skin cancer classification system, *IOP Conf. Ser., Mater Sci Eng*, 982, 1, Art. 012005.
- Barataand C, Marques JS (2019) Deep learning for skin cancer diagnosis with hierarchical architectures, in Proc. IEEE 16th Int. Symp. Biomed. Imag. (ISBI), pp. 841–845
- Nigar N, Umar M, Shahzad MK, Islam S, Abalo D (2022) A deep learning approach based on explainable artificial intelligence for skin lesion classification. *IEEE Access* 10:113715–113725
- Ameri A (2020) A deep learning approach to skin cancer detection in dermoscopy images, *J. Biomed. Phys. Eng* 10(6):801–806
- Kumar RS, Singh A, Srinath S, Thomas NK, Arasu V (2022) Skin cancer detection using deep learning, in Proc. Int. Conf. Electron. Renew. Syst. (ICEARS), Mar. pp. 1724–1730
- Nawaz M, Nazir T, Masood M, Ali F, Khan MA, Tariq U, Sahar N, Damaševičius R (2022) Melanoma segmentation: A framework of improved DenseNet77 and UNET convolutional neural network, *Int. J. Imag. Syst. Technol* 32(6):2137–2153
- Maqsood S, Damaševičius R (2023) Multiclass skin lesion localization and classification using deep learning-based features fusion and selection framework for smart healthcare, *Neural Netw* 160:238–258
- Abayomi-Alli OO, Damaševičius R, Misra S, Maskeliunas R, Abayomi-Alli A (2021) Malignant skin melanoma detection using image augmentation by oversampling in nonlinear lower-dimensional embedding manifold, *TURKISH J. Electr. Eng. Comput. Sci.* 29(SI-1):2600–2614

30. jiahao W, Xingguang J, Yuan W, Luo Z, Yu Z (2021) Deep neural network for melanoma classification in dermoscopic images, in Proc. IEEE Int. Conf. Consum. Electron. Comput. Eng. (ICCECE). pp. 666–669
31. Zhang Y, Wang C (2021) SIIM-ISIC melanoma classification with DenseNet, in Proc. IEEE 2nd Int. Conf. Big Data, Artif. Intell. Internet Things Eng. (ICBAIE). pp. 14–17
32. Zhang N, Cai Y-X, Wang Y-Y, Tian Y-T, Wang X-L, Badami B (2020) Skin cancer diagnosis based on optimized convolutional neural network, *artif. Intell Med*, 102, Art. 101756.
33. Liu Z, Xiong R, Jiang T (2023) CI-Net: Clinical-inspired network for automated skin lesion recognition, *IEEE Trans. Med. Imag* 42(3):619–632
34. Kaur R, GholamHosseini H, Sinha R, Lindén M (2022) Melanoma classification using a novel deep convolutional neural network with dermoscopic images, *sensors* 22(3):1134
35. Reis HC, Turk V, Khoshelham K, Kaya S (2022) InSiNet: A deep convolutional approach to skin cancer detection and segmentation, *Med. Biol. Eng. Comput* 60(3):643–662
36. López-Labraca J, González-Díaz I, Díaz-de-María F, Fueyo Casado A (2022) An interpretable CNN-based CAD system for skin lesion diagnosis, *artif. Intell Med*, 132, Art. 102370.
37. Rezk E, Eltorki M, El-Dakhakhni W (2023) Interpretable skin cancer classification based on incremental domain knowledge learning, *J. Healthcare Informat. Res.* 7(1):59–83
38. Wighton P, Lee TK, Lui H, McLean DI, Atkins MS (2011) Generalizing common tasks in automated skin lesion diagnosis, *IEEE Trans. Inf. Technol. Biomed* 15(4):622–629
39. Saeed J, Zeebaree S (2021) Skin lesion classification based on deep convolutional neural networks architectures, *J. Appl. Sci. Technol. Trends* 2(1):41–51
40. Li Y, Esteve A, Kuprel B, Novoa R, Ko J, Thrun S (2016) Skin cancer detection and tracking using data synthesis and deep learning, *arXiv:1612.01074*.
41. Jamil U, Akram MU, Khalid S, Abbas S, Saleem K (2016) Computer based melanocytic and nevus image enhancement and segmentation, *BioMed Res. Int* 2016:1–13
42. Lee YC, Jung S-H, Won H-H (2018) WonDerM: skin lesion classification with fine-tuned neural networks, *arXiv:1808.03426*.
43. Badrinarayanan V, Kendall A, Cipolla R (2017) SegNet: A deep convolutional encoder–decoder architecture for image segmentation, *IEEE Trans. Pattern Anal. Mach. Intell* 39(12):2481–2495
44. Hosny KM, Kassem MA, Foad MM (2018) Skin cancer classification using deep learning and transfer learning, in Proc. 9th Cairo Int. Biomed. Eng. Conf. (CIBEC). pp. 90–93
45. Javaid A, Sadiq M, Akram F (2021) Skin cancer classification using image processing and machine learning, in Proc. Int. Bhurban Conf. Appl. Sci. Technol. (IBCAST). pp. 439–444
46. Uckunerand M, Erol H (2021) A new deep learning model for skin cancer classification, in Proc. 6th Int. Conf. Comput. Sci. Eng. (UBMK). pp. 27–31
47. Filali Y, Khoukhi HE, Sabri MA, Aarab A (2022) Analysis and classification of skin cancer based on deep learning approach, in Proc. Int. Conf. Intell. Syst. Comput. Vis. (ISCV). pp. 1–6
48. Kaleli F, Aydin N, Ertas G, Gulcur HO (2007) An adaptive approach to the segmentation of DCE-MR images of the breast: Comparison with classical thresholding algorithms, in Proc. IEEE Symp. Comput. Intell. Image Signal Process. pp. 375–379
49. Saba T (2021) Computer vision for microscopic skin cancer diagnosis using handcrafted and non-handcrafted features, *Microsc. Res. Technique* 84(6):1272–1283
50. Rout R, Parida P (Feb. 2020) A novel method for melanocytic skin lesion extraction and analysis, *J. Discrete Math Sci Cryptogr* 23(2):461–473
51. Rout R, Priyadarsan P (2023) and D.Sonali, A hybrid deep learning network for skin lesion extraction, in Proc. Int. Conf. Soft Comput. Pattern Recognit pp. 682–689
52. Ferreira CA, Cunha A, Mendonça AM, Campilho A (2019) Convolutional neural network architectures for texture classification of pulmonary nodules. In: Vera-rodriguez R, Fierrez J, Morales A (eds) *Progress in pattern Recognition, image Analysis, computer Vision, and applications*. Springer, Cham, Switzerland 11401:783–791
53. Ali A-R, Li J, Yang G, O'Shea SJ (2020) A machine learning approach to automatic detection of irregularity in skin lesion border using dermo scopic images, *PeerJ Comput. Sci* 6:e268
54. Ali A-R, Li J, O'Shea SJ (2020) Towards the automatic detection of skin lesion shape asymmetry, color variegation and diameter in dermoscopic images, *PLoS One* 15(6):Art. no. e0234352.
55. Das R, De S, Bhattacharyya S, Platos J, Snael V, Hassanien AE (2020) Data augmentation and feature fusion for melanoma detection with content-based image classification, in Proc. Int. Conf. Adv. Mach. Learn. Technol. Appl., A. E. Hassanien, A. T. Azar, T. Gaber, R. Bhatnagar, and M. F. Tolba, Eds. Cham, Switzerland: Springer 921:712–721
56. Rehman A, Khan MA, Mehmood Z, Saba T, Sardaraz M, Rashid M (2020) Microscopic melanoma detection and classification: A framework of pixel-based fusion and multilevel features reduction, *Microsc. Res. Technique* 83(4):410–423
57. Rizwan W, Adnan SM, Ahmed W, Faizi MI (2021) Skin lesions detection and classification using deep learning, *Warse* 10(3):1720–1728
58. Codella NCF, Nguyen Q-B, Pankanti S, Gutman DA, Helba B, Halpern AC, Smith JR (2017) Deep learning ensembles for melanoma recognition in dermoscopy images, *IBM J. Res. Develop* 61(4–5):1–15
59. Barata C, Ruela M, Francisco M, Mendonça T, Marques JS (2014) Two systems for the detection of melanomas in dermoscopy images using texture and color features, *IEEE Syst. J.* 8(3):965–979
60. Wong A, Scharcanski J, Fieguth P (2011) Automatic skin lesion segmentation via iterative stochastic region merging, *IEEE Trans. Inf. Technol. Biomed* 15(6):929–936
61. Rubegni P, Cevenini G, Burroni M, Perotti R, Dell'Eva G, Sbrano P, Miracco C, Luzi P, Tosi P, Barbini P, Andreassi L (2002) Automated diagnosis of pigmented skin lesions, *Int. J. Cancer* 101(6):576–580
62. Dash S, Parida P, Mohanty JR (2023) Illumination robust deep Convolutional neural network for medical image classification. *Soft Comput.*
63. Rout R, Parida P, Alotaibi Y, Alghamdi S, Khalaf OI (2021) Skin lesion extraction using multiscale morphological local variance reconstruction based watershed transform and fast fuzzy c-means clustering, *Symmetry* 13(11):2085
64. Tan TY, Zhang L, Lim CP (2020) Adaptive melanoma diagnosis using evolving clustering, ensemble and deep neural networks. *Knowl-Based Syst*, 187, Art. 104807.
65. K wasigroch A, Grochowski M, Mikolajczyk A (2020) Neural architecture search for skin lesion classification. *IEEE Access* 8:9061–9071
66. Adegun AA, Viriri S (2020) Deep learning-based system for automatic melanoma detection. *IEEE Access* 8:7160–7172
67. Song L, Lin J, Wang ZJ, Wang H (2020) An end-to-end multi-task deep learning framework for skin lesion analysis, *IEEE J. Biomed. Health Informat* 24(10):2912–2921
68. Yu L, Chen H, Dou Q, Qin J, Heng P (2017) Automated melanoma recognition in dermoscopy images via very deep residual networks. *IEEE Trans Med Imag* 36(4):994–1004

69. DeVries T, Ramachandram D (2017) Skin lesion classification using deep multi-scale convolutional neural networks, arXiv:1703.01402.
70. Mahbod A, Schaefer G, Wang C, Ecker R, Elling I (2019) Skin lesion classification using hybrid deep neural networks, in Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP), pp. 1229–1233
71. Kalouche S, Ng A, Duchi J (2016) Vision-based classification of skin cancer using deep learning. Stanford Univ., Stanford, CA, USA, Tech. Rep.,
72. Hasan MK, Elahi MTE, Alam MA, Jawad MT, Martí R (2022) DermoExpert: skin lesion classification using a hybrid convolutional neural network through segmentation, transfer learning, and augmentation. *Informat Med Unlocked*, 28, Art. 100819.
73. Shorfuzzaman M (2022) An explainable stacked ensemble of deep learning models for improved melanoma skin cancer detection, *Multimedia Syst.* 28(4):1309–1323
74. Benyahia S, Meftah B, Lézoray O (2022) Multi-features extraction based on deep learning for skin lesion classification. *Tissue Cell*, 74, Art. 101701.
75. Agyenta C, Akanzawon M (2022) Skin lesion classification based on convolutional neural network, *J. Appl. Sci. Technol. Trends* 3(1):14–19
76. Bhosale S (2022) Comparison of deep learning and machine learning models and frameworks for skin lesion classification, arXiv:2207.12715.
77. Bechelli S, Delhommelle J (2022) Machine learning and deep learning algorithms for skin cancer classification from dermoscopic images, *Bioengineering* 9(3):97
78. Malo DC, Rahman MM, Mahbub J, Khan MM (2022) Skin cancer detection using convolutional neural network, in Proc. IEEE 12th Annu. Comput. Commun. Workshop Conf. (CCWC), pp. 0169–0176
79. Afza F, Sharif M, Khan MA, Tariq U, Yong H-S, Cha J (2022) Multiclass skin lesion classification using hybrid deep features selection and extreme learning machine, *sensors*. 22(3):799
80. Popescu D, El-khatib M, Ichim L (2022) Skin lesion classification using collective intelligence of multiple neural networks, *Sensors* 22(12):4399
81. Singha S, Roy P (2022) Skin cancer classification and comparison of pretrained models' performance using transfer learning, pp. 1–8
82. Keerthana D, Venugopal V, Nath MK, Mishra M (2023) Hybrid convolutional neural networks with SVM classifier for classification of skin cancer, *biomed. Eng Adv*, 5, Art. 100069.
83. Jain S, Singhania U, Tripathy B, Nasr EA, Aboudaif MK, Kamrani AK (2021) Deep learning-based transfer learning for classification of skin cancer, *Sensors* 21(23):8142
84. Zhang Z, Li Y, Wang F (2024) Enhanced deep information fusion based on residual U-Net (ED-ResUNet). *Applied intelligence*. Springer
85. Liu Z, Chen R, Gao X (2025) MRP-UNet: Multiscale input fusion U-Net for skin lesion segmentation, *Scientific Reports*, vol. –, pp. –, Nature Publishing Group
86. Sethanan K, Somboon N, Jitpech S (2023) Double AMIS-ensemble deep learning for skin cancer classification, *Expert Systems with Applications* 232, art. no. 120895
87. Hosseinzadeh M, Ghaemi R, Akbari M (2024) A deep ensemble model for skin cancer detection and classification using HAM10000 dataset, *Journal of King Saud University – Computer and Information Sciences*, vol. –, pp. –, Elsevier
88. Chiu TM, Huang YC, Lee CC (2025) Deep ensemble learning for multiclass skin lesion classification, *Computers in Biology and Medicine* 173, art. no. 108092
89. Li J (2022) A survey on firefly algorithms: theory and applications. *Neurocomputing* 505:58–75
90. Thomas E, George A, Mathew J (2024) Fuzzy C-Means clustering coupled with firefly optimization for medical image segmentation. *Procedia Comput Sci* 234:345–353
91. <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:0.7910/DVN/DBW86T>
92. Kassem MA, Hosny KM, Fouad MM (2020) Skin lesions classification into eight classes for ISIC 2019 using deep convolutional neural network and transfer learning. *IEEE Access* 8:114822–114832
93. abud-allah H (2023) A review of approaches and difficulties in the use of deep learning techniques for COVID-19 disease diagnosis on chest X-Rays. *EDRAAK* 33–38. <https://doi.org/10.70470/EDRAAK/2023/007>
94. Alobaid A, Bonny T A (2024) Comparative Analysis of Machine and Deep Learning Models in the Early Detection of Breast Cancer, 2024 Advances in Science and Engineering Technology International Conferences (ASET), Abu Dhabi, United Arab Emirates, 2024:1–9. <https://doi.org/10.1109/ASET60340.2024.10708703>
95. Abud-Allah H (2024) Multi-view Deep Learning-Based COVID-19 Diagnosis with Chest X-Ray Images: A Comparative Study of SVM and KNN Classifiers. *EDRAAK* 2024:59–77. <https://doi.org/10.70470/EDRAAK/2024/009>

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